



ENGINE SPECIFICATION • CLINICAL COVERAGE MATRIX • EVIDENCE & IP
BASIS

ARKA Imaging Intelligence Engine AIIE — Technical Compendium

A complete, source-grounded reference to how AIIE derives imaging appropriateness — its scoring methodology, the deterministic Clinical Knowledge Matrix, and the recommended modality for every clinical variant and patient discriminator the engine resolves.

DOCUMENT
AIIE Technical Compendium

PREPARED
June 10, 2026

ENGINE / MATRIX VERSION
Engine 1.0 • Knowledge Matrix v1.0.0

CONFIDENTIAL • INTERNAL & INVESTOR REVIEW

Notices, Scope & Honest-Use Statement

This compendium documents a working clinical-decision-support engine. It is written to be defensible under scrutiny — by clinicians, by investors, and by regulators — which means it states plainly what AIIE is, what it is not, and what has and has not yet been measured.

WHAT AIIE IS

AIIE (the ARKA Imaging Intelligence Engine) is a transparent, **non-device clinical decision support** function. For a proposed imaging order it returns an appropriateness score on a 1–9 scale together with the individual factors, weights, and primary-literature citations behind that score. It is designed to meet the four criteria for Non-Device CDS under section 520(o)(1)(E) of the U.S. Food, Drug & Cosmetic Act: it does not acquire or analyse image pixel data; it operates on structured medical information; it offers a recommendation to a licensed clinician; and it exposes its full basis so the clinician can independently review and override it.

WHAT AIIE IS NOT — AND MUST NOT BE PRESENTED AS

AIIE is **not** an autonomous diagnostic device, is **not** a substitute for clinical judgment, and is **not** claimed to be clinically superior to any published appropriateness standard. The ratings in this document are a **prototype knowledge base**. The engine's measured performance to date is **74% three-class agreement on a synthetic, appropriateness-aligned validation cohort**; real-world diagnostic-accuracy and AUC figures are **pending pilot data and are not asserted here**. No statement in this document should be read as a head-to-head superiority claim over the ACR Appropriateness Criteria[®] or any other guideline.

INDEPENDENT DERIVATION & INTELLECTUAL-PROPERTY BASIS

AIIE's ratings are **independently derived** from primary, openly citable evidence — RAND/UCLA appropriateness methodology, GRADE evidence-to-decision framing, Choosing Wisely, USPSTF, ACEP, PECARN, the Ottawa and Canadian/NEXUS decision rules, BEIR/ICRP radiation guidance, AAST injury scoring, ACS and society screening guidance, TI-RADS, and ASHA practice guidance, among others. AIIE does **not** reproduce, transcribe, or redistribute the copyrighted text of the ACR Appropriateness Criteria[®]. ARKA, AIIE, the ARKA Imaging Intelligence Engine, and "remARKAbly precise" are marks of ARKA Health. **ARKA is not affiliated with, sponsored by, or endorsed by the American College of Radiology**. Part III sets out the IP and positioning rationale in full.

NOT MEDICAL, LEGAL, OR FINANCIAL ADVICE

This document is engineering and regulatory-positioning reference material. It is not medical advice and must not be used to direct the care of any individual patient outside a validated, supervised deployment. It is not legal advice; the IP and regulatory analysis in Part III is informational and is pending review and sign-off by qualified counsel (see the project's `SCOPE_BOUNDARY.md`, counsel sign-off: pending).

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Document Control & How to Read This Compendium

Document title	AIIE Technical Compendium — Engine Specification, Clinical Coverage Matrix & Evidence/IP Basis
Engine	ARKA Imaging Intelligence Engine (AIIE) — shared scoring core for ARKA-CLIN and ARKA-INS
Knowledge Matrix version	v1.0.0 (deterministic, source-stamped into every score)
Source of truth	ARKA monorepo — <code>lib/aiie/scoring-engine.ts</code> , <code>lib/aiie/knowledge-matrix/</code>
Prepared	June 10, 2026
Classification	Confidential — internal, clinical advisory, and investor review
Intended audience	Clinical advisors, regulatory/IP counsel, engineering, and prospective investors/partners
Regulatory posture	Non-Device CDS, FD&C Act §520(o)(1)(E); transparency-by-design

At a glance

14 BODY REGIONS	36 CLINICAL SCENARIOS	93 CLINICAL VARIANTS	14 MODALITIES	1,302 MODALITY RATINGS
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Every one of the 1,302 ratings in Part II is generated directly from the engine's source data, not hand-authored for this document. 16 variants encode a deliberate *imaging-not-indicated* recommendation — the engine is built to discourage low-value imaging as readily as it endorses high-value imaging.

How this document is organised

- **Part I — Engine Specification.** How AIIE turns an order into a 1–9 score: the scale, the six weighted factors plus the trauma gate, the deterministic four-tier Knowledge Matrix, the normalizer, the safety gates, payer-facing mapping (MNAI / denial-risk), confidence, the optional ML regressor, and validation.
- **Part II — Clinical Coverage Matrix.** Region by region, scenario by scenario, variant by variant: the recommended modality and AIIE score for each clinical situation and patient discriminator, with per-row rationale, radiation burden, and an evidence anchor.
- **Part III — Evidence & IP Basis.** How AIIE relates to the ACR Appropriateness Criteria[®] without reproducing them, and how to describe AIIE's performance without overclaiming.
- **Appendices.** Scale legend, modality and radiation dictionary, red-flag dictionary, the primary-evidence register, region-default (Tier 3) tables, methodology and change control, and a glossary.

READING THE SCORE CHIPS

Throughout Part II each modality carries a colour-coded AIIE score: **2** 1–3 usually not appropriate · **5** 4–6 may be appropriate · **8** 7–9 usually appropriate. A ★ marks the engine's preferred first-line study for that variant.

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PART I

Engine Specification

How AIIE converts a single proposed imaging order into a transparent 1–9 appropriateness score — the scale, the weighted factor model, the deterministic Clinical Knowledge Matrix, the normalizer that accepts any input, the safety gates, the payer-facing mapping, the confidence model, and the optional machine-learning layer.

IN THIS PART

1. I.1 Architecture overview
2. I.2 The AIIE 1–9 appropriateness scale
3. I.3 Scoring model: factors, weights, SHAP-style deltas
4. I.4 The Clinical Knowledge Matrix
5. I.5 Deterministic resolution — the four-tier cascade
6. I.6 Order-context normalization
7. I.7 Safety gates — trauma, STAT, swallow
8. I.8 Demographic & contextual modifiers
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10. I.10 Payer-facing layer — MNAI & denial-risk
11. I.11 Optional ML regressor & graceful degradation
12. I.12 Validation posture, limitations & honest metrics
13. I.13 Auditability & reproducibility

I.1 Architecture overview

AIIE is the shared scoring brain behind two ARKA products: **ARKA-CLIN**, which advises a clinician at imaging order-entry, and **ARKA-INS**, which expresses the same clinical signal in payer/prior-authorization terms.

Both call one function, `scoreOrder()`, so a patient and a payer are evaluated against an identical, auditable standard.

The methodology is stated explicitly in the engine header: a **RAND/UCLA appropriateness** framing with **GRADE-style** evidence-to-decision transparency. Every factor emits a bounded, SHAP-style delta from a neutral baseline, so the final score is always decomposable into named, cited contributions rather than an opaque number.

Processing is deterministic and local. A proposed order is normalized into a structured context, resolved against the Clinical Knowledge Matrix, scored across the weighted factors, passed through the relevant safety gates, and returned with the per-factor breakdown, a plain-language narrative, a confidence estimate, and a matrix-version stamp. No pixel data is read and no network call is required to reach a recommendation.

Stage	What happens
1 • Normalize	<code>normalizeOrderContext()</code> maps any input — structured FHIR fields, CPT/ICD-10 codes, or free text — into a <code>NormalizedOrderContext</code> . Never throws.
2 • Resolve	<code>resolveRating()</code> walks the four-tier matrix cascade to an evidence-linked modality rating and a match tier.
3 • Score	Six weighted factors (plus the trauma gate) each emit a bounded delta; the sum moves the score off the neutral baseline of 5.
4 • Gate	Trauma-severity, STAT-priority, and swallow-study triage gates apply structured-data overrides where relevant.
5 • Map & explain	Clinical 1–9 score, payer denial-risk proxy, MNAI policy alignment, confidence, SHAP-style factors, and narrative are assembled into the response.

TWO NUMBERS, ONE SOURCE OF TRUTH

The clinical appropriateness score (1–9, higher = more appropriate) and the payer denial-risk proxy (1–9, higher = more likely to be denied) are mirror images of the same computation — the denial-risk proxy is literally `10 - clinic-alScore`. A clinician and a payer never see two different opinions dressed up as one.

I.2 The AIIE 1–9 appropriateness scale

AIIE expresses appropriateness on a nine-point ordinal scale anchored at a neutral midpoint of **5**. The scale is partitioned into three clinician-facing bands. ARKA labels every value an **"AIIE score"**; it is the engine's own output and is never labelled an "ACR rating."

1 – 3

Usually not appropriate

Imaging unlikely to benefit; may add radiation, cost, or false-positive harm.

4 – 6

May be appropriate

Conditional; depends on documentation, sequence, or shared decision-making.

7 – 9

Usually appropriate

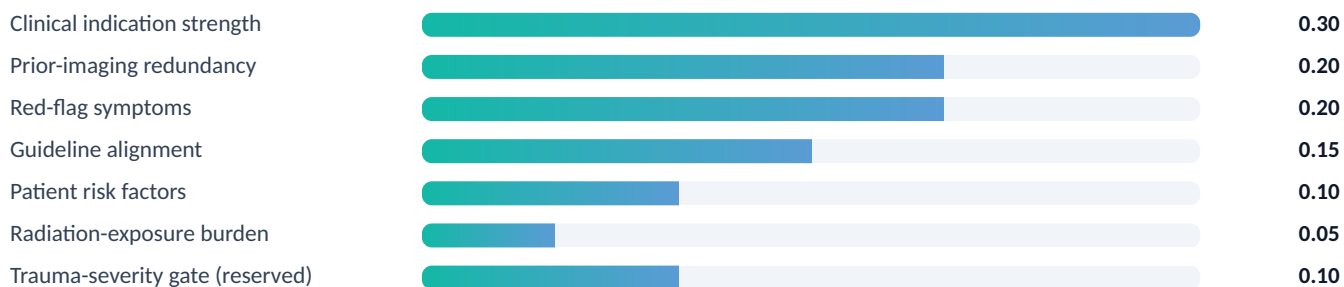
Well-supported first-line or confirmatory study for the presentation.

The resolved matrix rating maps linearly onto a signed signal around the midpoint: a rating of 9 corresponds to a maximally supportive signal of +1, a rating of 1 to -1, and 5 to 0. Formally the indication signal is $(\text{rating} - 5) / 4$, clamped to the range -1...+1. The same linear mapping recurs wherever the matrix rating informs a factor.

I.3 Scoring model: factors, weights, and SHAP-style deltas

Six weighted factors describe an ordinary order; a seventh — the trauma-severity gate — is reserved and activates only when structured trauma scores are present. Each factor computes a signal in -1...+1, then contributes $\text{weight} \times \text{signal} \times 4$ points to the score. The constant 4 (the delta-scale) is chosen so that fully aligned signals move the score by roughly ± 4 from the baseline of 5 before rounding.

DEFAULT FACTOR WEIGHTS



Weights sum to 1.00. When the trauma gate is active it holds 0.10 and the remaining six factors are rescaled to share 0.90, preserving the bounded-delta property.

FINAL ASSEMBLY

`clinicalScore = clamp( round( 5 +  $\sum(\text{weight} \times \text{signal} \times 4)$  ), 1, 9 )`. The factors are then sorted by absolute contribution to produce a natural-language narrative naming the top supportive and tempering drivers, and the contributions are surfaced verbatim on the CDS card so the clinician sees exactly why the score landed where it did.

I.3.1 Clinical indication strength

Default weight: 0.30

The strongest single factor. It reads the resolved matrix rating through the linear map $(\text{rating}-5)/4$ and adds a bounded delta from the live record snapshot (active problems, coded indications). It answers: does the evidence base support this study for this presentation?

EVIDENCE ANCHOR

GRADE summary-of-findings framing for symptom-test concordance; RAND/UCLA indication panels.

I.3.2 Prior-imaging redundancy

Default weight: 0.20

Penalises repeat imaging of the same region when a recent prior study exists and conservative management has not intervened. A recent prior without conservative care drives the signal sharply negative (-0.85); an older prior, or one followed by appropriate conservative care, is penalised little or not at all.

EVIDENCE ANCHOR

Choosing Wisely and appropriateness guidance on redundant imaging in stable presentations.

I.3.3 Red-flag symptoms

Default weight: 0.20

Escalates appropriateness when dangerous features are present. It scales with the number of active red flags (normalised against six) and adds a severity boost for neurologic deficit, bladder/bowel dysfunction, or cancer history — the features most associated with time-critical diagnoses.

EVIDENCE ANCHOR

ACEP and specialty red-flag guidance for urgent diagnoses requiring expedited imaging.

I.3.4 Guideline alignment

Default weight: 0.15

A second, independent read of the matrix rating, attenuated by how specifically the matrix matched the order. An exact-variant match carries full weight; scenario, region, and conservative defaults carry 0.85, 0.60, and 0.35 respectively, so the engine is explicitly less confident when it had to fall back.

EVIDENCE ANCHOR

AIIE Clinical Knowledge Matrix rating with match-tier confidence attenuation.

I.3.5 Patient risk factors

Default weight: 0.10

Population risk modifiers: extremes of age (≥ 65 or ≤ 5), pregnancy, and immunosuppression each raise the signal, reflecting altered pre-test probability and the heightened stakes of missed diagnosis in these groups.

EVIDENCE ANCHOR

Consensus screening and risk-stratification guidance (age, pregnancy, immune status).

I.3.6 Radiation-exposure burden

Default weight: 0.05

A small but deliberate counterweight. Ionising, high-dose studies (CT, PET, nuclear medicine) receive a negative signal; radiation-free studies (MRI, ultrasound) a slight positive one — encouraging the lowest-dose study that answers the clinical question, all else equal.

EVIDENCE ANCHOR

BEIR / ICRP-informed dose trade-offs for ionising versus non-ionising modalities.

I.3.7 Trauma-severity gate (reserved 0.10)

Default weight: —

Active only when the record snapshot carries structured trauma scores (GCS, ISS, AIS, RTS, REMS via their LOINC codes). It translates measured injury severity into an order-specific signal and, when present, reserves 0.10 of the total weight while the other six factors are rescaled to 0.90.

EVIDENCE ANCHOR

AAST ISS/AIS injury severity and ACR head-trauma criteria (GCS, REMS); structured EHR trauma observations.

I.3.8 Score assembly, rounding, and the narrative

Contributions are summed, added to the baseline of 5, rounded to the nearest integer, and clamped to 1–9. Because each contribution is bounded and signed, the result is always explainable as a short list of push-up and push-down forces. The engine emits a sentence of the form: *"Overall appropriateness maps to N/9 on the RAND/UCLA-style scale. Supportive drivers include Factors tempering appropriateness include Contributions are reported as transparent, GRADE-aligned deltas from a neutral baseline of 5."*

I.4 The Clinical Knowledge Matrix

The matrix is the deterministic clinical knowledge base that gives AIIE its content. It is owned by ARKA, versioned (`v1.0.0`), and stamped into every score for audit. It is organised as a strict hierarchy.

Level	Definition
Body region	14 anatomic partitions (head/brain, spine segments, chest, cardiac, abdomen, pelvis, GU/renal, MSK upper & lower, vascular, breast, head/face/neck).
Clinical scenario	36 presentation clusters (e.g. Acute Headache, Low Back Pain, RLQ Pain/Appendicitis). Each carries presentation keywords, ICD-10 prefixes, and optional SNOMED codes for matching.
Scenario variant	93 discriminated branches. A variant's criteria select it by red flags, symptom duration, age range, pregnancy, trauma, prior imaging, immune status, and whether conservative care was tried.
Modality rating	For each variant, all 14 modalities receive a 1–9 rating, a radiation tier (0–4), an evidence slug, a clinician-facing rationale, and an optional preferred flag and contrast caution.
Evidence slug	A stable kebab-case identifier shared by every modality in a variant, linking the rating set to its evidence metadata and enabling regression testing.

The matrix therefore enumerates **1,302** modality ratings (93 variants × 14 modalities), each independently citable. Part II prints every one of them.

IMAGING RESTRAINT IS FIRST-CLASS

16 of the 93 variants are explicitly flagged `imagingIndicated: false` — encoding situations (uncomplicated acute low back pain, minor head injury meeting low-risk rules, routine pre-operative clearance) where the most appropriate action is *no imaging*. The engine recommends restraint as confidently as it recommends a study.

I.5 Deterministic resolution: the four-tier cascade

Resolution is strict, ordered, and total — it never throws and never returns null. Any input lands in exactly one of four tiers, and the tier is recorded so downstream factors can attenuate their confidence.

Tier	Confidence	When it fires
1 — <code>exact_variant</code>	1.00	The first ranked scenario candidate that has a fully matching variant. Red-flag variants outrank non-red-flag variants; ties break on the number of specified criteria, then stable file order.
2 — <code>scenario_default</code>	0.85	A scenario matched but no variant criteria fit; the scenario's default variant is used.
3 — <code>region_default</code>	0.60	Region and modality are known but no scenario matched; a deliberately conservative mid-scale rating is returned with a documentation-improvement prompt.
4 — <code>conservative_default</code>	0.35	Nothing resolvable; a neutral rating of 4 is returned with explicit guidance on what to document (complaint, region, modality, duration, red-flag status, prior workup) to obtain a definitive rating.

Within the matched variant the engine returns the rating for the requested modality; if the modality is unknown it falls back to the variant's preferred study, or failing that the highest-rated one. This is why AIIE can always answer "what should I have ordered instead?" — the preferred modality is intrinsic to every variant.

I.6 Order-context normalization

The normalizer is what lets AIIE accept messy, real-world input without failing. It maps structured FHIR fields, imaging CPT codes, ICD-10 indications, and free clinical text into a single normalized context, deterministically and without network calls. Common 70000-series radiology CPT subranges resolve to a modality and often a region; a duration parser converts phrases such as "3 weeks" into days for the variant duration windows; presentation keywords rank scenario candidates; and structured trauma observations are read straight from the snapshot. Unmapped or garbage input still produces a valid context — it simply resolves to a lower tier.

DURATION PARSING

The duration parser is deterministic and returns a normalized value in weeks with a tempo pattern (constant, intermittent, progressive, improving, or unknown) and a confidence level. It recognises clinical abbreviations ("3 wks", "2 mos") and ranges and maps them onto the day-based windows the matrix variants use — for example separating acute low back pain under six weeks from radiculopathy persisting beyond it.

CPT TO MODALITY / REGION MAPPING

When an imaging CPT code is present the normalizer resolves it directly to a modality and, where the range is anatomically specific, a body region — anchoring resolution even when free text is sparse. The complete mapping the engine uses is reproduced in Appendix F.

I.7 Safety gates: trauma, STAT, and swallow triage

I.7.1 Trauma-severity gate

Reads GCS, ISS, AIS panel, RTS, and REMS observations (by LOINC code) from the record snapshot and converts measured injury severity into a bounded order-specific signal ($-0.8...+0.8$). Scores are treated as measurements, not diagnoses, consistent with AAST injury-severity resources and ACR head-injury criteria.

I.7.2 STAT-priority gate

Restricts a true STAT designation to genuinely emergent criteria — $GCS \leq 13$, suspected acute stroke (NIHSS-aligned lexicon), suspected PE with haemodynamic instability, suspected aortic dissection, severe trauma ($ISS \geq 16$ or instability), and unstable pediatric fever — and otherwise recommends an Urgent (4-hour) priority, protecting STAT pathways from overuse.

I.7.3 Swallow-study triage

Selects among a video-fluoroscopic swallow study (VFSS), fiberoptic endoscopic evaluation (FEES), and bedside clinical evaluation based on the clinical context (stroke, post-extubation, progressive neuromuscular disease, esophageal-phase concern, head-and-neck cancer), citing ASHA practice guidance and dysphagia appropriateness evidence.

I.8 Demographic and contextual modifiers

AIIE does not treat demographics as cosmetic. They enter the score through three concrete channels, which is why the same complaint yields different recommendations for, say, a pregnant patient versus a non-pregnant one, or a child versus an older adult.

Discriminator	How it moves the recommendation
Age	Selects age-banded variants (e.g. pediatric ultrasound-first vs adult CT-first for appendicitis) and raises the patient-risk signal at the extremes (≥ 65 or ≤ 5).
Sex & pregnancy	Routes pregnancy-sensitive scenarios to non-ionising first-line studies (ultrasound/MRI for suspected appendicitis, renal colic, or ectopic) and raises patient risk; contrast cautions are attached at the modality level.
Symptom duration	Drives duration-windowed variants — the difference between imaging an acute presentation and one that has persisted past a conservative-care threshold (e.g. radiculopathy > 6 weeks).
Prior imaging & conservative care	Engage the redundancy factor and unlock 'after conservative care' variants, so the engine rewards an appropriate work-up sequence rather than immediate advanced imaging.
Red flags	Select the most specific high-acuity variant and raise the red-flag signal, escalating both the score and, where relevant, the priority.

I.9 Confidence model

Each score carries a confidence estimate beginning at 0.62 and rising as the order is better documented — present symptoms, a substantive chief complaint, a stated prior-imaging timeframe, a conservative-management duration, a CPT code, and known payer each add a small increment, capped at 0.95. Confidence reflects completeness of the input, not certainty of the diagnosis, and is shown alongside the score rather than baked into it.

I.10 Payer-facing layer: MNAI and denial-risk mapping

For ARKA-INS the clinical score is expressed in authorization terms without changing the clinical numerics. The **denial-risk proxy** is `10 - clinicalScore`, and a routing function assigns a handling band: gold-carded providers auto-approve; denial-risk ≤ 3 is likely-approve; ≤ 6 routes to clinical review; higher is likely-deny.

The **Medical Necessity Alignment Index (MNAI)** is a separate, deterministic check that matches the order's ICD-10/CPT pair against curated payer-policy tables and returns a 0–100 alignment index and a green/amber/red tier with the supporting policy references. MNAI retrieves evidence; it does *not* alter the AIIE clinical score.

I.11 The optional ML regressor and graceful degradation

Alongside the deterministic engine, ARKA maintains an optional gradient-boosted regressor (the ARKA Imaging Appropriateness Regressor, XGBoost, v1.0.0) trained on a 23-feature vector. It is a convenience layer, not the source of truth: when an order falls outside the model's distribution the system degrades gracefully to the deterministic rules-plus-citations output rather than failing closed.

Attribute	Value
Type	XGBoost gradient-boosted regressor, output 1–9 binned to low / uncertain / high
Features	23 structured features (no pixel data, no free-text embeddings)
Training data	5,000 synthetic, appropriateness-aligned examples (70/15/15 split); labels derived from rule-based scenarios plus Gaussian noise ($\sigma \approx 0.3$); contains no PHI
Reported (synthetic test set)	RMSE ≈ 1.41 ; three-class accuracy $\approx 73.9\%$ (74%); MAE and R^2 reported per run
Planned	AUC, calibration slope, and expected calibration error — for retrospective validation when labelled real-world outcomes exist
Out of scope	Time-critical alerts, image interpretation, autonomous ordering, inpatient-only pathways, and screening populations outside the trained indication mix

I.12 Validation posture, limitations, and honest metrics

STATE THE MEASURED NUMBER, AND ONLY THAT

AIIE's validated metric today is **74% three-class accuracy on a synthetic, appropriateness-aligned validation cohort** (RMSE ≈ 1.41). This is an investor-demo-grade figure on synthetic data, **not** a real-world clinical-performance claim. Real-world AUC and diagnostic accuracy are **pending pilot data**. AIIE makes **no** claim of superiority over the ACR Appropriateness Criteria[®] or any other published standard; no head-to-head study has been run.

Known limitations follow directly from the design: the knowledge base is a curated prototype covering 36 scenarios, not the full breadth of clinical medicine; the ML layer is trained on synthetic data and its subgroup fairness across age, sex, and modality requires confirmation on real cohorts; and out-of-distribution orders intentionally degrade to conservative, documentation-seeking output. These are stated so that no reader mistakes a prototype for a finished clinical product.

I.13 Auditability and reproducibility

Determinism is a feature. Given the same input, AIIE returns the same score, the same factor breakdown, and the same matrix-version stamp every time — a prerequisite for clinical sign-off, regulatory review, and regression testing. The matrix enforces integrity at load time (unique scenario IDs, unique and internally consistent evidence slugs), and the rules path runs without ML so a reviewer can trace any recommendation end to end. Transparency of this kind is also what keeps AIIE within the Non-Device CDS definition: the clinician can independently review the basis of every recommendation.

PART II

Clinical Coverage Matrix

The recommended imaging for every clinical situation AIIE resolves. 14 regions, 36 scenarios, 93 variants, and all 1,302 modality ratings — each with an AIIE 1–9 score, radiation burden, rationale, and evidence anchor, generated directly from the engine.

IN THIS PART

1. How to read this part
2. One section per body region
3. Every variant shows all 14 modalities
4. Patient discriminators (age, sex, pregnancy, duration, red flags) shown per variant

II.0 How to read the coverage matrix

Each region below contains its clinical scenarios; each scenario contains one or more variants; each variant is a specific patient situation. For every variant, all fourteen modalities are listed with their AIIE score so the recommended study and the discouraged ones are both explicit.

- The **Applies when** line states the exact discriminators — red flags, duration window, age range, pregnancy, trauma, prior imaging, and conservative-care status — that select the variant.
- Modalities are sorted most- to least-appropriate. The ★ marks AIIE's preferred first-line study.
- Rows greyed at AIIE 1 are modalities the engine considers not indicated for that situation; they are shown for completeness so every combination is accounted for.
- A **IMAGING NOT INDICATED** tag marks variants where the most appropriate action is no imaging at all.

INDEPENDENT DERIVATION

These ratings are ARKA's own, derived from the primary evidence registered in Appendix D. They are aligned to the same three-band clinical logic the field uses, but they are not reproduced from, and are not a copy of, the ACR Appropriateness Criteria[®]. See Part III.

II.1 Head / Brain

4 scenarios · 17 variants in this region.

Neuroimaging triage turns on tempo and deficit. AIIE separates hyperacute presentations — thunderclap headache, suspected stroke — that demand emergent non-contrast CT to exclude haemorrhage before any intervention, from chronic stable headache where imaging is usually unnecessary. MRI is favoured where soft-tissue, posterior-fossa, or infective detail and radiation avoidance matter; CT leads where speed and haemorrhage detection dominate. Pediatric and minor-injury variants apply validated low-risk rules to keep children away from avoidable radiation.

Acute Headache

Region: Head / Brain · ICD-10: [R51](#) , [G44](#)

New or worsening headache requiring neuroimaging triage. Across 7 variants, AIIE's preferred first-line studies CT, MRI, MRI (C+), XR. Imaging is explicitly discouraged in 2 variants (low-value-care guard).

Thunderclap

thunderclap

Applies when: red flags: sudden / hyperacute onset.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	9	Moderate	Non-contrast head CT is first-line for thunderclap headache to exclude subarachnoid hemorrhage.
CTA CT angiography	7	Moderate	CTA may follow non-contrast CT when aneurysm or vascular cause is suspected.
MRA MR angiography	6	None	MRA can supplement CTA for aneurysm evaluation when CT is inconclusive.
MRI MRI, non-contrast	5	None	MRI is less available emergently and may miss acute SAH compared with CT.
MRI (C+) MRI with contrast	4	None	Contrast MRI is not first-line for acute thunderclap headache workup.
CT (C+) CT with IV contrast	4	Moderate	Contrast CT adds limited value acutely for SAH detection compared with non-contrast CT.
US Ultrasound	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
XR Radiography (plain film)	1	Minimal	

MODALITY	AIIE	RADIATION	RATIONALE
			Not indicated for this neurologic presentation; alternative modalities are preferred.
Mammo Mammography	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
DEXA DEXA densitometry	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Fluoro Fluoroscopy	1	Low	Not indicated for this neurologic presentation; alternative modalities are preferred.
NM Nuclear medicine	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
PET-CT PET-CT	1	High	Not indicated for this neurologic presentation; alternative modalities are preferred.

Evidence anchor: [head-brain-acute-headache-thunderclap](#)

Neuro Deficit

neuro-deficit

Applies when: red flags: focal neurologic deficit.

MODALITY	AIIE	RADIATION	RATIONALE
MRI ★ MRI, non-contrast	9	None	MRI brain is preferred for headache with focal neurologic deficit to evaluate stroke, mass, or demyelination.
CT CT, non-contrast	8	Moderate	Non-contrast CT is appropriate when MRI is unavailable or contraindicated.
MRA MR angiography	7	None	MRA helps evaluate vascular etiologies when deficit suggests ischemia or dissection.
CTA CT angiography	7	Moderate	CTA is useful when acute vascular pathology is suspected alongside neurologic deficit.
MRI (C+) MRI with contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Ultrasound	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
XR Radiography (plain film)	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Mammo Mammography	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
DEXA DEXA densitometry	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Fluoro		Low	

MODALITY	AIIE	RADIATION	RATIONALE
Fluoroscopy	1		Not indicated for this neurologic presentation; alternative modalities are preferred.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
NM Nuclear medicine	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
PET-CT PET-CT	1	High	Not indicated for this neurologic presentation; alternative modalities are preferred.

Evidence anchor: [head-brain-acute-headache-neuro-deficit](#)

Fever Immunocompromised

fever-immunocompromised

Applies when: red flags: fever / infectious concern, immunocompromised host.

MODALITY	AIIE	RADIATION	RATIONALE
MRI (C+) ★ MRI with contrast	9	None	Contrast-enhanced MRI is preferred for headache with fever in immunocompromised hosts to detect CNS infection or abscess.
CT (C+) CT with IV contrast	7	Moderate	Contrast CT is an alternative when MRI is unavailable in suspected CNS infection.
MRI MRI, non-contrast	6	None	Non-contrast MRI may miss meningeal or parenchymal enhancement in infection.
MRA MR angiography	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Ultrasound	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
XR Radiography (plain film)	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Mammo Mammography	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
DEXA DEXA densitometry	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Fluoro Fluoroscopy	1	Low	Not indicated for this neurologic presentation; alternative modalities are preferred.
CT CT, non-contrast	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
CTA CT angiography	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.

NM

MODALITY	AIIE	RADIATION	RATIONALE
Nuclear medicine	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
PET-CT PET-CT	1	High	Not indicated for this neurologic presentation; alternative modalities are preferred.

Evidence anchor: [head-brain-acute-headache-fever-immunocompromised](#)

SCENARIO DEFAULT

IMAGING NOT INDICATED

Chronic Stable

chronic-stable

Applies when: duration ≥ 42 d (6+ wk).

MODALITY	AIIE	RADIATION	RATIONALE
MRI MRI, non-contrast	3	None	MRI may be considered only if red flags emerge during conservative management.
CT CT, non-contrast	2	Moderate	Routine CT is not indicated for chronic stable headache without red flags per Choosing Wisely.
MRI (C+) MRI with contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
MRA MR angiography	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Ultrasound	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
XR ★ Radiography (plain film)	1	Minimal	Skull radiographs do not evaluate intracranial causes of chronic stable headache.
Mammo Mammography	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
DEXA DEXA densitometry	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Fluoro Fluoroscopy	1	Low	Not indicated for this neurologic presentation; alternative modalities are preferred.
CTA CT angiography	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
NM Nuclear medicine	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
PET-CT PET-CT	1	High	Not indicated for this neurologic presentation; alternative modalities are preferred.

Evidence anchor: [head-brain-acute-headache-chronic-stable](#)**New over 50**

new-over-50

Applies when: red flags: age > 50.

MODALITY	AIIE	RADIATION	RATIONALE
MRI ★ MRI, non-contrast	8	None	New headache after age 50 warrants MRI to exclude secondary causes including mass and giant cell arteritis territory disease.
CT CT, non-contrast	7	Moderate	CT is acceptable when MRI is delayed or contraindicated for new headache over 50.
MRI (C+) MRI with contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
MRA MR angiography	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Ultrasound	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
XR Radiography (plain film)	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Mammo Mammography	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
DEXA DEXA densitometry	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Fluoro Fluoroscopy	1	Low	Not indicated for this neurologic presentation; alternative modalities are preferred.
CTA CT angiography	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
NM Nuclear medicine	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
PET-CT PET-CT	1	High	Not indicated for this neurologic presentation; alternative modalities are preferred.

Evidence anchor: [head-brain-acute-headache-new-over-50](#)

Post Trauma GCS under 15

post-trauma-gcs-under-15

Applies when: red flags: recent trauma; acute trauma.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	9	Moderate	Non-contrast head CT is indicated for head trauma with GCS less than 15 per major trauma guidelines.
MRI MRI, non-contrast	5	None	MRI is not first-line acutely for moderate-severe traumatic brain injury evaluation.
MRI (C+) MRI with contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
MRA MR angiography	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Ultrasound	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
XR Radiography (plain film)	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Mammo Mammography	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
DEXA DEXA densitometry	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Fluoro Fluoroscopy	1	Low	Not indicated for this neurologic presentation; alternative modalities are preferred.
CTA CT angiography	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
NM Nuclear medicine	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
PET-CT PET-CT	1	High	Not indicated for this neurologic presentation; alternative modalities are preferred.

Evidence anchor: [head-brain-acute-headache-post-trauma-gcs-under-15](#)

IMAGING NOT INDICATED Post Trauma GCS 15 Low Risk

post-trauma-gcs-15-low-risk

Applies when: red flags: recent trauma; acute trauma; after conservative care.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	2	Moderate	

MODALITY	AIIE	RADIATION	RATIONALE
			CT is usually not appropriate for minor head trauma meeting validated low-risk clinical decision rule criteria.
MRI MRI, non-contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
MRI (C+) MRI with contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
MRA MR angiography	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Ultrasound	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
XR Radiography (plain film)	1	Minimal	Skull radiographs do not assess intracranial injury after head trauma.
Mammo Mammography	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
DEXA DEXA densitometry	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Fluoro Fluoroscopy	1	Low	Not indicated for this neurologic presentation; alternative modalities are preferred.
CTA CT angiography	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
NM Nuclear medicine	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
PET-CT PET-CT	1	High	Not indicated for this neurologic presentation; alternative modalities are preferred.

Evidence anchor: [head-brain-acute-headache-post-trauma-gcs-15-low-risk](#)

Suspected Stroke

Region: Head / Brain · ICD-10: [I63](#) , [I64](#) , [G45](#)

Acute cerebrovascular event evaluation for thrombolysis or intervention. Across 3 variants, AIIE's preferred first-line studies CT, MRI.

SCENARIO DEFAULT **Acute under 4 5h**

acute-under-4-5h

Applies when: red flags: sudden / hyperacute onset; duration ≤ 1 d.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	9	Moderate	Non-contrast head CT is mandatory emergently to exclude hemorrhage before thrombolysis.
CTA CT angiography	9	Moderate	CTA head/neck identifies large-vessel occlusion guiding thrombectomy candidacy.
MRI MRI, non-contrast	8	None	MRI with diffusion is highly sensitive for acute ischemia when CT is negative.
MRA MR angiography	8	None	MRA complements CTA for intracranial vessel assessment in acute stroke.
MRI (C+) MRI with contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Ultrasound	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
XR Radiography (plain film)	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Mammo Mammography	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
DEXA DEXA densitometry	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Fluoro Fluoroscopy	1	Low	Not indicated for this neurologic presentation; alternative modalities are preferred.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
NM Nuclear medicine	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
PET-CT PET-CT	1	High	Not indicated for this neurologic presentation; alternative modalities are preferred.

Evidence anchor: [head-brain-suspected-stroke-acute-under-4-5h](#)**Wake Up Stroke**

wake-up-stroke

Applies when: red flags: sudden / hyperacute onset.

MODALITY	AIIE	RADIATION	RATIONALE
MRI ★ MRI, non-contrast	9	None	

MODALITY	AIIE	RADIATION	RATIONALE
			MRI with DWI is preferred for wake-up stroke to define infarct age and eligibility for intervention.
CT CT, non-contrast	8	Moderate	Non-contrast CT excludes hemorrhage as the initial safety study.
CTA CT angiography	8	Moderate	CTA evaluates large-vessel occlusion in wake-up stroke protocols.
MRI (C+) MRI with contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
MRA MR angiography	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Ultrasound	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
XR Radiography (plain film)	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Mammo Mammography	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
DEXA DEXA densitometry	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Fluoro Fluoroscopy	1	Low	Not indicated for this neurologic presentation; alternative modalities are preferred.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
NM Nuclear medicine	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
PET-CT PET-CT	1	High	Not indicated for this neurologic presentation; alternative modalities are preferred.

Evidence anchor: [head-brain-suspected-stroke-wake-up](#)

TIA

tia

Applies when: red flags: sudden / hyperacute onset; duration ≤ 1 d.

MODALITY	AIIE	RADIATION	RATIONALE
MRI ★ MRI, non-contrast	9	None	MRI brain is recommended for TIA to detect acute infarction and guide secondary prevention.
MRA MR angiography	8	None	MRA neck evaluates carotid disease in TIA patients without immediate CTA need.
CTA	8	Moderate	CTA or MRA of carotids is appropriate in TIA workup for stenosis.

MODALITY	AIIE	RADIATION	RATIONALE
CT angiography			
CT CT, non-contrast	6	Moderate	CT alone may miss small cortical infarcts in TIA.
MRI (C+) MRI with contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Ultrasound	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
XR Radiography (plain film)	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Mammo Mammography	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
DEXA DEXA densitometry	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Fluoro Fluoroscopy	1	Low	Not indicated for this neurologic presentation; alternative modalities are preferred.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
NM Nuclear medicine	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
PET-CT PET-CT	1	High	Not indicated for this neurologic presentation; alternative modalities are preferred.

Evidence anchor: [head-brain-suspected-stroke-tia](#)

Seizure

Region: Head / Brain · ICD-10: [R56](#) , [G40](#)

New or breakthrough seizure requiring neuroimaging triage. Across 2 variants, AIIE's preferred first-line study MRI.

SCENARIO DEFAULT First Seizure

first-seizure

Applies when: applies as the scenario baseline (no additional discriminators).

MODALITY	AIIE	RADIATION	RATIONALE
MRI ★ MRI, non-contrast	9	None	MRI brain is appropriate for first unprovoked seizure in adults to identify structural lesions.
CT CT, non-contrast	7	Moderate	CT is acceptable emergently when MRI is unavailable or patient is unstable.
MRI (C+) MRI with contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.

MODALITY	AIIE	RADIATION	RATIONALE
MRA MR angiography	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Ultrasound	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
XR Radiography (plain film)	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Mammo Mammography	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
DEXA DEXA densitometry	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Fluoro Fluoroscopy	1	Low	Not indicated for this neurologic presentation; alternative modalities are preferred.
CTA CT angiography	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
NM Nuclear medicine	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
PET-CT PET-CT	1	High	Not indicated for this neurologic presentation; alternative modalities are preferred.

Evidence anchor: [head-brain-seizure-first](#)

Breakthrough

breakthrough

Applies when: red flags: progressive symptoms.

MODALITY	AIIE	RADIATION	RATIONALE
MRI ★ MRI, non-contrast	8	None	Breakthrough seizure with new features warrants repeat MRI to exclude progressive structural disease.
CT CT, non-contrast	6	Moderate	CT may be used when MRI access is limited in breakthrough seizure evaluation.
MRI (C+) MRI with contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
MRA MR angiography	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Ultrasound	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.

MODALITY	AIIE	RADIATION	RATIONALE
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
XR Radiography (plain film)	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Mammo Mammography	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
DEXA DEXA densitometry	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Fluoro Fluoroscopy	1	Low	Not indicated for this neurologic presentation; alternative modalities are preferred.
CTA CT angiography	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
NM Nuclear medicine	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
PET-CT PET-CT	1	High	Not indicated for this neurologic presentation; alternative modalities are preferred.

Evidence anchor: [head-brain-seizure-breakthrough](#)

Head Trauma

Region: Head / Brain · ICD-10: [S06](#) , [S09](#)

Traumatic head injury stratified by age and clinical decision rules. Across 5 variants, AIIE's preferred first-line study CT. Imaging is explicitly discouraged in 2 variants (low-value-care guard).

SCENARIO DEFAULT Adult High Risk

adult-high-risk

Applies when: red flags: recent trauma, focal neurologic deficit; acute trauma.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	9	Moderate	CT head is indicated for adult head trauma with high-risk features per validated decision rules.
MRI MRI, non-contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
MRI (C+) MRI with contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
MRA MR angiography	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Ultrasound	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.

MODALITY	AIIE	RADIATION	RATIONALE
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
XR Radiography (plain film)	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Mammo Mammography	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
DEXA DEXA densitometry	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Fluoro Fluoroscopy	1	Low	Not indicated for this neurologic presentation; alternative modalities are preferred.
CTA CT angiography	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
NM Nuclear medicine	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
PET-CT PET-CT	1	High	Not indicated for this neurologic presentation; alternative modalities are preferred.

Evidence anchor: [head-brain-head-trauma-adult-high-risk](#)

IMAGING NOT INDICATED Adult Low Risk

adult-low-risk

Applies when: red flags: recent trauma; age ≥ 18 y; acute trauma.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	2	Moderate	CT is usually not appropriate when adult minor head injury meets low-risk clinical criteria.
MRI MRI, non-contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
MRI (C+) MRI with contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
MRA MR angiography	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Ultrasound	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
XR Radiography (plain film)	1	Minimal	Skull radiographs do not assess intracranial traumatic injury.
Mammo		Minimal	

MODALITY	AIIE	RADIATION	RATIONALE
Mammography	1		Not indicated for this neurologic presentation; alternative modalities are preferred.
DEXA DEXA densitometry	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Fluoro Fluoroscopy	1	Low	Not indicated for this neurologic presentation; alternative modalities are preferred.
CTA CT angiography	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
NM Nuclear medicine	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
PET-CT PET-CT	1	High	Not indicated for this neurologic presentation; alternative modalities are preferred.

Evidence anchor: [head-brain-head-trauma-adult-low-risk](#)

Pediatric under 2

pediatric-under-2

Applies when: red flags: recent trauma, age < 18; age ≤ 2 y; acute trauma.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	8	Moderate	CT head is often indicated in infants under 2 with head trauma due to limited exam and abuse concern.
MRI MRI, non-contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
MRI (C+) MRI with contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
MRA MR angiography	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Ultrasound	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
XR Radiography (plain film)	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Mammo Mammography	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
DEXA DEXA densitometry	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Fluoro		Low	

MODALITY	AIIE	RADIATION	RATIONALE
Fluoroscopy	1		Not indicated for this neurologic presentation; alternative modalities are preferred.
CTA CT angiography	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
NM Nuclear medicine	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
PET-CT PET-CT	1	High	Not indicated for this neurologic presentation; alternative modalities are preferred.

Evidence anchor: [head-brain-head-trauma-pediatric-under-2](#)

Pediatric 2 to 18

pediatric-2-to-18

Applies when: red flags: recent trauma, age < 18; age 2–18 y; acute trauma.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	7	Moderate	CT is indicated when pediatric head trauma fails PECARN low-risk criteria.
MRI MRI, non-contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
MRI (C+) MRI with contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
MRA MR angiography	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Ultrasound	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
XR Radiography (plain film)	1	Minimal	Skull films are not appropriate for pediatric intracranial injury assessment.
Mammo Mammography	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
DEXA DEXA densitometry	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Fluoro Fluoroscopy	1	Low	Not indicated for this neurologic presentation; alternative modalities are preferred.
CTA CT angiography	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
CT (C+)	1	Moderate	

MODALITY	AIIE	RADIATION	RATIONALE
CT with IV contrast			Not indicated for this neurologic presentation; alternative modalities are preferred.
NM Nuclear medicine	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
PET-CT PET-CT	1	High	Not indicated for this neurologic presentation; alternative modalities are preferred.

Evidence anchor: [head-brain-head-trauma-pediatric-2-to-18](#)

IMAGING NOT INDICATED

Pediatric Low Risk

pediatric-low-risk

Applies when: red flags: recent trauma, age < 18; age 2–18 y; acute trauma; after conservative care.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	2	Moderate	CT is usually not appropriate when pediatric head trauma meets PECARN low-risk criteria; observation avoids unnecessary radiation.
MRI MRI, non-contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
MRI (C+) MRI with contrast	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
MRA MR angiography	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Ultrasound	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this neurologic presentation; alternative modalities are preferred.
XR Radiography (plain film)	1	Minimal	Skull radiographs do not assess intracranial injury in pediatric head trauma.
Mammo Mammography	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
DEXA DEXA densitometry	1	Minimal	Not indicated for this neurologic presentation; alternative modalities are preferred.
Fluoro Fluoroscopy	1	Low	Not indicated for this neurologic presentation; alternative modalities are preferred.
CTA CT angiography	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
NM Nuclear medicine	1	Moderate	Not indicated for this neurologic presentation; alternative modalities are preferred.
PET-CT		High	

MODALITY	AIIE	RADIATION	RATIONALE
PET-CT	1		Not indicated for this neurologic presentation; alternative modalities are preferred.

Evidence anchor: [head-brain-head-trauma-pediatric-low-risk](#)

II.2 Head, Face & Neck

3 scenarios · 5 variants in this region.

Most uncomplicated sinusitis is a clinical diagnosis and is not imaged; CT is reserved for suspected orbital or intracranial complications. Thyroid nodules are characterised first by ultrasound with structured risk stratification, and persistent or red-flag adult neck masses are worked up with contrast-enhanced CT to map nodal and primary disease.

Sinusitis

Region: Head, Face & Neck · ICD-10: [J01](#) , [J32](#)

Acute and complicated sinusitis imaging triage. Across 2 variants, AIIE's preferred first-line study CT. Imaging is explicitly discouraged in 1 variant (low-value-care guard).

SCENARIO DEFAULT

IMAGING NOT INDICATED

Uncomplicated

uncomplicated

Applies when: applies as the scenario baseline (no additional discriminators).

MODALITY	AIIE	RADIATION	RATIONALE
MRI MRI, non-contrast	1	None	MRI is not indicated for routine uncomplicated sinusitis.
MRI (C+) MRI with contrast	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Sinus radiographs do not change management of uncomplicated acute sinusitis.
Mammo Mammography	1	Minimal	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
CT ★ CT, non-contrast	1	Moderate	Imaging is usually not appropriate for uncomplicated acute sinusitis diagnosed clinically.
CTA CT angiography	1	Moderate	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.

MODALITY	AIIE	RADIATION	RATIONALE
NM Nuclear medicine	1	Moderate	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.

Evidence anchor: [head-face-neck-sinusitis-uncomplicated](#)

Complicated Orbital

complicated-orbital

Applies when: red flags: fever / infectious concern, focal neurologic deficit.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	8	Moderate	CT sinuses with contrast is appropriate for complicated sinusitis with orbital or intracranial extension concern.
CT (C+) CT with IV contrast	8	Moderate	Contrast CT evaluates orbital cellulitis, abscess, and cavernous sinus involvement.
MRI MRI, non-contrast	7	None	MRI supplements CT for intracranial complications of sinusitis.
MRI (C+) MRI with contrast	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.

Evidence anchor: [head-face-neck-sinusitis-complicated-orbital](#)

Thyroid Nodule

Region: Head, Face & Neck · ICD-10: [E04](#) , [D34](#)

Thyroid nodule characterization and FNA triage. Across 1 variant, AIIE's preferred first-line study US.

SCENARIO DEFAULT

Initial Characterization

initial-characterization

Applies when: applies as the scenario baseline (no additional discriminators).

MODALITY	AIIE	RADIATION	RATIONALE
US ★ Ultrasound	9	None	Thyroid ultrasound is first-line for nodule characterization and TI-RADS assessment.
NM Nuclear medicine	5	Moderate	Thyroid scintigraphy is for hyperfunctioning nodule assessment, not all nodules.
CT CT, non-contrast	3	Moderate	CT is not first-line for thyroid nodule evaluation and delivers neck radiation.
MRI MRI, non-contrast	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.

Evidence anchor: [head-face-neck-thyroid-nodule-initial-characterization](#)

Neck Mass (Adult)

Region: Head, Face & Neck · ICD-10: [R22](#) , [R59](#)

Adult neck mass evaluation including malignancy workup. Across 2 variants, AIIE's preferred first-line studies CT, CT (C+).

SCENARIO DEFAULT Persistent Adult

persistent-adult

Applies when: duration ≥ 14 d (2+ wk); age ≥ 18 y.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	8	Moderate	Contrast CT neck is appropriate for persistent adult neck mass to evaluate nodal and primary sites.
CT (C+) CT with IV contrast	8	Moderate	Contrast-enhanced CT defines neck mass extent and vascular involvement.
MRI MRI, non-contrast	7	None	MRI neck is an alternative for soft-tissue characterization without radiation.
US Ultrasound	6	None	Ultrasound may characterize superficial nodes but CT/MRI evaluates deep spaces.
PET-CT PET-CT	6	High	PET-CT may stage known malignancy but is not first-line for initial neck mass.
MRI (C+) MRI with contrast	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.

Evidence anchor: [head-face-neck-neck-mass-adult-persistent](#)

With Red Flags

with-red-flags

Applies when: red flags: unintentional weight loss, prior cancer history, fever / infectious concern.

MODALITY	AIIE	RADIATION	RATIONALE
CT (C+) ★ CT with IV contrast	9	Moderate	Contrast CT neck is indicated for neck mass with systemic red flags to exclude malignancy or abscess.
PET-CT PET-CT	7	High	PET-CT may aid oncologic staging when malignancy is confirmed.
MRI MRI, non-contrast	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this head, face, or neck presentation; preferred modalities address the clinical question.

Evidence anchor: [head-face-neck-neck-mass-adult-with-red-flags](#)

II.3 Cervical Spine

1 scenario · 4 variants in this region.

Atraumatic mechanical neck pain without neurologic signs is usually not imaged. AIIE escalates to MRI when myelopathy or progressive deficit is suspected, and to CT when blunt cervical trauma fails a validated low-risk clinical clearance rule — while explicitly discouraging imaging when those rules are satisfied.

Neck Pain

Region: Cervical Spine · ICD-10: [M54.2](#) , [M50](#) , [S13](#)

Cervical spine pain including myelopathy and trauma triage. Across 4 variants, AIIE's preferred first-line studies XR, MRI, CT. Imaging is explicitly discouraged in 2 variants (low-value-care guard).

SCENARIO DEFAULT

IMAGING NOT INDICATED

Uncomplicated

uncomplicated

Applies when: atraumatic.

MODALITY	AIIE	RADIATION	RATIONALE
MRI MRI, non-contrast	2	None	MRI is not indicated for uncomplicated neck pain without red flags or persistent symptoms.
MRI (C+) MRI with contrast	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
XR ★ Radiography (plain film)	1	Minimal	Imaging is usually not appropriate for uncomplicated atraumatic neck pain without neurologic deficit.
Mammo Mammography	1	Minimal	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	CT cervical spine is not appropriate for uncomplicated mechanical neck pain.
CTA CT angiography	1	Moderate	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.

MODALITY	AIIE	RADIATION	RATIONALE
NM Nuclear medicine	1	Moderate	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.

Evidence anchor: [spine-cervical-neck-pain-uncomplicated](#)

Myelopathy Signs

myelopathy-signs

Applies when: red flags: focal neurologic deficit, progressive symptoms.

MODALITY	AIIE	RADIATION	RATIONALE
MRI ★ MRI, non-contrast	9	None	MRI cervical spine is indicated for suspected myelopathy to evaluate cord compression.
CT CT, non-contrast	4	Moderate	CT does not adequately assess spinal cord or disc herniation causing myelopathy.
MRI (C+) MRI with contrast	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.

Evidence anchor: [spine-cervical-neck-pain-myelopathy-signs](#)**Trauma High Risk**

trauma-high-risk

Applies when: red flags: recent trauma; acute trauma.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	9	Moderate	CT cervical spine is indicated when cervical trauma fails NEXUS or Canadian C-spine low-risk criteria.
XR Radiography (plain film)	5	Minimal	Radiographs may be used in very low-risk trauma but CT is preferred when criteria are not met.
MRI MRI, non-contrast	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.

Evidence anchor: [spine-cervical-neck-pain-trauma-high-risk](#)**IMAGING NOT INDICATED Trauma Low Risk**

trauma-low-risk

Applies when: red flags: recent trauma; acute trauma; after conservative care.

MODALITY	AIIE	RADIATION	RATIONALE
XR			

MODALITY	AIIE	RADIATION	RATIONALE
Radiography (plain film)	2	Minimal	Cervical radiographs are usually not needed when clinical clearance criteria are satisfied.
CT ★ CT, non-contrast	2	Moderate	CT is usually not appropriate when blunt cervical trauma meets validated low-risk clinical decision criteria.
MRI MRI, non-contrast	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this cervical spine presentation; preferred modalities address the clinical question.

Evidence anchor: [spine-cervical-neck-pain-trauma-low-risk](#)

II.4 Thoracic Spine

1 scenario · 3 variants in this region.

Uncomplicated atraumatic mid-back pain is rarely imaged. Systemic red flags — known malignancy, fever, weight loss — route to MRI to exclude metastatic or infective disease, and significant trauma routes to CT.

Thoracic Back Pain

Region: Thoracic Spine · ICD-10: [M54.6](#) , [M51](#) , [S22](#)

Mid-back pain triage including red flags and trauma. Across 3 variants, AIIE's preferred first-line studies XR, MRI, CT. Imaging is explicitly discouraged in 1 variant (low-value-care guard).

SCENARIO DEFAULT

IMAGING NOT INDICATED

Uncomplicated

uncomplicated

Applies when: atraumatic.

MODALITY	AIIE	RADIATION	RATIONALE
MRI MRI, non-contrast	2	None	MRI is not first-line for uncomplicated thoracic mechanical back pain.
MRI (C+) MRI with contrast	1	None	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
XR ★ Radiography (plain film)	1	Minimal	Imaging is usually not appropriate for uncomplicated atraumatic thoracic back pain without red flags.
Mammo Mammography	1	Minimal	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	

MODALITY	AIIE	RADIATION	RATIONALE
			Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.

Evidence anchor: [spine-thoracic-thoracic-back-pain-uncomplicated](#)

Red Flags

red-flags

Applies when: red flags: prior cancer history, fever / infectious concern, unintentional weight loss.

MODALITY	AIIE	RADIATION	RATIONALE
MRI ★ MRI, non-contrast	9	None	MRI thoracic spine is indicated for back pain with systemic red flags to exclude malignancy or infection.
CT CT, non-contrast	7	Moderate	CT may evaluate osseous pathology when MRI is contraindicated.
MRI (C+) MRI with contrast	1	None	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.

Evidence anchor: [spine-thoracic-thoracic-back-pain-red-flags](#)

Post Trauma

post-trauma

Applies when: red flags: recent trauma; acute trauma.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	8	Moderate	CT thoracic spine is appropriate for significant thoracic trauma with pain or neurologic symptoms.
XR Radiography (plain film)	6	Minimal	Thoracic radiographs may screen for fracture but CT is more sensitive.
MRI MRI, non-contrast	1	None	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this thoracic spine presentation; preferred modalities address the clinical question.

Evidence anchor: [spine-thoracic-thoracic-back-pain-post-trauma](#)

II.5 Lumbar Spine

1 scenario · 7 variants in this region.

The lumbar spine is the engine's clearest demonstration of low-value-care reduction: uncomplicated acute low back pain under six weeks without red flags is explicitly not imaged. Advanced imaging is unlocked only by red flags (suspected cauda equina, infection, or malignancy) or by radiculopathy persisting past a conservative-care threshold — with MRI as the workhorse and CT or radiographs in defined secondary roles.

Low Back Pain

Region: Lumbar Spine · ICD-10: [M54](#) , [M51](#) , [M48](#)

Lumbar spine pain triage including red flags and duration-based imaging. Across 7 variants, AIIE's preferred first-line studies XR, MRI, MRI (C+), CT. Imaging is explicitly discouraged in 1 variant (low-value-care guard).

SCENARIO DEFAULT

IMAGING NOT INDICATED

Uncomplicated under 6wk

uncomplicated-under-6wk

Applies when: duration ≤ 42 d (6 wk); no conservative care yet.

MODALITY	AIIE	RADIATION	RATIONALE
MRI MRI, non-contrast	1	None	MRI is usually not appropriate for uncomplicated acute low back pain under 6 weeks per ACR Appropriateness Criteria.
MRI (C+) MRI with contrast	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
XR ★ Radiography (plain film)	1	Minimal	Imaging is usually not appropriate for uncomplicated acute low back pain under 6 weeks without red flags.
Mammo Mammography	1	Minimal	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	CT lumbar spine is not indicated for uncomplicated acute low back pain without red flags.
CTA CT angiography	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	

MODALITY	AIIE	RADIATION	RATIONALE
			Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.

Evidence anchor: [spine-lumbar-low-back-pain-uncomplicated-under-6wk](#)

Radiculopathy over 6wk

radiculopathy-over-6wk

Applies when: duration ≥ 42 d (6+ wk); after conservative care.

MODALITY	AIIE	RADIATION	RATIONALE
MRI ★ MRI, non-contrast	8	None	MRI lumbar spine is appropriate for persistent radiculopathy after 6 weeks of conservative care.
CT CT, non-contrast	4	Moderate	CT is a secondary option when MRI is contraindicated for radiculopathy evaluation.
XR Radiography (plain film)	3	Minimal	Lumbar radiographs have limited value for radiculopathy after failed conservative therapy.
MRI (C+) MRI with contrast	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
PET-CT		High	

MODALITY	AIIE	RADIATION	RATIONALE
PET-CT	1		Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.

Evidence anchor: [spine-lumbar-low-back-pain-radiculopathy-over-6wk](#)

Cauda Equina

cauda-equina

Applies when: red flags: bladder/bowel dysfunction, focal neurologic deficit.

MODALITY	AIIE	RADIATION	RATIONALE
MRI ★ MRI, non-contrast	9	None	Emergent MRI lumbar spine is indicated for suspected cauda equina syndrome — expedite.
CT CT, non-contrast	5	Moderate	CT may be used only when MRI is unavailable in cauda equina suspicion.
MRI (C+) MRI with contrast	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.

Evidence anchor: [spine-lumbar-low-back-pain-cauda-equina](#)

Cancer History

cancer-history

Applies when: red flags: prior cancer history.

MODALITY	AIIE	RADIATION	RATIONALE
MRI ★ MRI, non-contrast	9	None	MRI is indicated for back pain with cancer history to evaluate metastatic or epidural disease.
CT CT, non-contrast	7	Moderate	CT may supplement MRI for osseous metastatic assessment when MRI is limited.
PET-CT PET-CT	6	High	PET-CT may stage known malignancy but is not first-line for focal back pain workup.
MRI (C+) MRI with contrast	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.

Evidence anchor: [spine-lumbar-low-back-pain-cancer-history](#)

Infection IVDU

infection-ivdu

Applies when: red flags: fever / infectious concern, IV drug use.

MODALITY	AIIE	RADIATION	RATIONALE
MRI (C+) ★ MRI with contrast	9	None	

MODALITY	AIIE	RADIATION	RATIONALE
			Contrast-enhanced MRI is preferred for suspected spinal infection or epidural abscess, especially with IVDU.
CT (C+) CT with IV contrast	6	Moderate	Contrast CT is inferior to MRI for early discitis and epidural infection.
MRI MRI, non-contrast	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.

Evidence anchor: [spine-lumbar-low-back-pain-infection-ivdu](#)

Osteoporosis Fragility

osteoporosis-fragility

Applies when: red flags: osteoporosis / fragility, recent trauma.

MODALITY	AIIE	RADIATION	RATIONALE
MRI ★ MRI, non-contrast	9	None	MRI detects occult osteoporotic compression fractures and cord compromise not seen on radiographs.
DEXA DEXA densitometry	7	Minimal	DEXA assesses bone density after fragility fracture but does not localize acute injury.
XR		Minimal	

MODALITY	AIIE	RADIATION	RATIONALE
Radiography (plain film)	6		Lumbar radiographs may show compression fracture but miss early marrow edema.
MRI (C+) MRI with contrast	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.

Evidence anchor: [spine-lumbar-low-back-pain-osteoporosis-fragility](#)

Post Trauma

post-trauma

Applies when: red flags: recent trauma; acute trauma.

MODALITY	AIIE	RADIATION	RATIONALE
MRI MRI, non-contrast	8	None	MRI is indicated when neurologic deficit or ligamentous injury is suspected after trauma.
CT ★ CT, non-contrast	8	Moderate	CT lumbar spine is appropriate for significant trauma when fracture is suspected.
XR Radiography (plain film)	6	Minimal	Radiographs may screen for fracture but CT or MRI is preferred for high-energy trauma.
MRI (C+) MRI with contrast	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
MRA		None	

MODALITY	AIIE	RADIATION	RATIONALE
MR angiography	1		Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this lumbar spine presentation; preferred modalities address the clinical question.

Evidence anchor: [spine-lumbar-low-back-pain-post-trauma](#)

II.6 Chest

4 scenarios · 11 variants in this region.

Acute chest pain is triaged across the three life-threats — acute coronary syndrome, pulmonary embolism, and aortic dissection — with pretest probability and D-dimer gating CT pulmonary angiography so low-probability patients are not over-imaged. Chronic cough is radiograph-first, and low-dose CT screening is confined to the eligible high-risk smoking population.

Acute Chest Pain

Region: Chest · ICD-10: [R07](#) , [I20](#) , [I26](#) , [I71](#)

Emergency chest pain triage for ACS, PE, and aortic pathology. Across 5 variants, AIIE's preferred first-line studies CTA, CT. Imaging is explicitly discouraged in 1 variant (low-value-care guard).

SCENARIO DEFAULT

Suspected ACS

suspected-acs

Applies when: red flags: sudden / hyperacute onset.

MODALITY	AIIE	RADIATION	RATIONALE
CTA ★ CT angiography	8	Moderate	Coronary CTA is appropriate for low-intermediate risk chest pain in selected ACS rule-out pathways.
NM Nuclear medicine	8	Moderate	Nuclear stress imaging is appropriate for intermediate-risk chest pain when CCTA is not suitable.
CT CT, non-contrast	7	Moderate	Chest CT may evaluate alternate diagnoses when ACS is not the sole concern.
XR Radiography (plain film)	4	Minimal	Chest radiograph alone is insufficient for ACS evaluation.
MRI MRI, non-contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	

MODALITY	AIIE	RADIATION	RATIONALE
			Not indicated for this chest presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this chest presentation; preferred modalities address the clinical question.

Evidence anchor: [chest-acute-chest-pain-suspected-acs](#)

IMAGING NOT INDICATED PE Low Pretest D Dimer

pe-low-pretest-d-dimer

Applies when: after conservative care.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	2	Moderate	CT pulmonary angiography is usually not appropriate when PE pretest probability is low and D-dimer is negative.
CTA CT angiography	2	Moderate	CTA is not indicated for PE when clinical probability is low with negative D-dimer.
MRI MRI, non-contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this chest presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
PET-CT		High	

MODALITY	AIIE	RADIATION	RATIONALE
PET-CT	1		Not indicated for this chest presentation; preferred modalities address the clinical question.

Evidence anchor: [chest-acute-chest-pain-pe-low-pretest-d-dimer](#)

PE Intermediate Pretest

pe-intermediate-pretest

Applies when: red flags: sudden / hyperacute onset.

MODALITY	AIIE	RADIATION	RATIONALE
CT CT, non-contrast	8	Moderate	CTPA is the standard for PE diagnosis with intermediate clinical suspicion.
CTA ★ CT angiography	8	Moderate	CT pulmonary angiography is appropriate for intermediate pretest probability PE when D-dimer is elevated.
US Doppler Ultrasound, Doppler/duplex	5	None	Lower-extremity Doppler may support PE diagnosis but does not replace CTPA.
MRI MRI, non-contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this chest presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this chest presentation; preferred modalities address the clinical question.

Evidence anchor: [chest-acute-chest-pain-pe-intermediate-pretest](#)

PE High Pretest

pe-high-pretest

Applies when: red flags: sudden / hyperacute onset, focal neurologic deficit.

MODALITY	AIIE	RADIATION	RATIONALE
CT CT, non-contrast	9	Moderate	CT pulmonary angiography is first-line for high-probability PE.
CTA ★ CT angiography	9	Moderate	CTPA is indicated for high pretest probability pulmonary embolism — expedite when hemodynamically unstable.
MRI MRI, non-contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this chest presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this chest presentation; preferred modalities address the clinical question.

Evidence anchor: [chest-acute-chest-pain-pe-high-pretest](#)

Aortic Dissection

aortic-dissection

Applies when: red flags: sudden / hyperacute onset, focal neurologic deficit.

MODALITY	AIIE	RADIATION	RATIONALE
CTA ★ CT angiography	9	Moderate	CTA chest/abdomen/pelvis is first-line for suspected acute aortic dissection.

MODALITY	AIIE	RADIATION	RATIONALE
CT (C+) CT with IV contrast	9	Moderate	Contrast-enhanced CT angiography defines dissection flap and branch involvement.
MRA MR angiography	8	None	MRA is an alternative when iodinated contrast is contraindicated in dissection workup.
MRI MRI, non-contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this chest presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this chest presentation; preferred modalities address the clinical question.

Evidence anchor: [chest-acute-chest-pain-aortic-dissection](#)

Chronic Cough

Region: Chest · ICD-10: [R05](#)

Persistent cough beyond 8 weeks requiring chest imaging triage. Across 2 variants, AIIE's preferred first-line studies XR, CT.

SCENARIO DEFAULT Standard Workup

standard-workup

Applies when: duration ≥ 56 d (8+ wk).

MODALITY	AIIE	RADIATION	RATIONALE
XR ★ Radiography (plain film)	7	Minimal	Chest radiograph is appropriate initial imaging for chronic cough after clinical evaluation.

CT

MODALITY	AIIE	RADIATION	RATIONALE
CT, non-contrast	6	Moderate	Chest CT may follow non-diagnostic radiograph or when malignancy risk is elevated.
MRI MRI, non-contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this chest presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this chest presentation; preferred modalities address the clinical question.

Evidence anchor: [chest-chronic-cough-standard-workup](#)

With Hemoptysis Weight Loss

with-hemoptysis-weight-loss

Applies when: red flags: unintentional weight loss.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	8	Moderate	Chest CT is indicated for chronic cough with weight loss or concerning features for malignancy.
XR Radiography (plain film)	6	Minimal	Radiograph may be initial but CT is preferred when malignancy is suspected.
PET-CT PET-CT	5	High	PET-CT is for staging known malignancy, not initial chronic cough workup.
MRI		None	

MODALITY	AIIE	RADIATION	RATIONALE
MRI, non-contrast	1		Not indicated for this chest presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this chest presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.

Evidence anchor: [chest-chronic-cough-with-hemoptysis-weight-loss](#)

Hemoptysis

Region: Chest · ICD-10: [R04.2](#)

Coughing blood requiring airway and parenchymal evaluation. Across 2 variants, AIIE's preferred first-line studies CT, CTA.

SCENARIO DEFAULT Minor

minor

Applies when: applies as the scenario baseline (no additional discriminators).

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	8	Moderate	Chest CT is appropriate for hemoptysis to identify bronchial and parenchymal sources.
CTA CT angiography	7	Moderate	CTA may evaluate vascular causes when CT suggests bronchial artery pathology.
XR Radiography (plain film)	6	Minimal	Chest radiograph may be initial but CT is more sensitive for hemoptysis etiology.
MRI		None	

MODALITY	AIIE	RADIATION	RATIONALE
MRI, non-contrast	1		Not indicated for this chest presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this chest presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this chest presentation; preferred modalities address the clinical question.

Evidence anchor: [chest-hemoptysis-minor](#)

Massive

massive

Applies when: red flags: sudden / hyperacute onset.

MODALITY	AIIE	RADIATION	RATIONALE
CTA ★ CT angiography	9	Moderate	CTA chest is indicated for massive hemoptysis to localize bleeding source before intervention.
CT CT, non-contrast	8	Moderate	Chest CT with angiographic technique guides bronchial artery embolization planning.
MRI MRI, non-contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US		None	

MODALITY	AIIE	RADIATION	RATIONALE
Ultrasound	1		Not indicated for this chest presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this chest presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this chest presentation; preferred modalities address the clinical question.

Evidence anchor: [chest-hemoptysis-massive](#)

Lung Cancer Screening

Region: Chest · ICD-10: [Z12.2](#) , [Z87.891](#)

Low-dose CT screening for high-risk asymptomatic adults. Across 2 variants, AIIE's preferred first-line study CT. Imaging is explicitly discouraged in 1 variant (low-value-care guard).

SCENARIO DEFAULT Eligible 50 80 20 Pack Years

[eligible-50-80-20-pack-years](#)

Applies when: age 50–80 y.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	9	Moderate	Annual low-dose chest CT is appropriate for eligible adults age 50–80 with ≥20 pack-year smoking history.
XR Radiography (plain film)	3	Minimal	Chest radiograph is insufficient for lung cancer screening per USPSTF and ACR guidance.
PET-CT PET-CT	2	High	PET-CT is not a screening modality for average-risk lung cancer screening.
MRI MRI, non-contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.

MODALITY	AIIE	RADIATION	RATIONALE
MRA MR angiography	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this chest presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.

Evidence anchor: [chest-lung-cancer-screening-eligible-50-80-20-pack-years](#)

IMAGING NOT INDICATED Not Eligible

not-eligible

Applies when: age 18–49 y.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	2	Moderate	Low-dose CT screening is usually not appropriate outside USPSTF age and smoking criteria.
MRI MRI, non-contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this chest presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Chest radiograph is not indicated for lung cancer screening in ineligible patients.

MODALITY	AIIE	RADIATION	RATIONALE
Mammo Mammography	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this chest presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this chest presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this chest presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this chest presentation; preferred modalities address the clinical question.

Evidence anchor: [chest-lung-cancer-screening-not-eligible](#)

II.7 Cardiac

3 scenarios · 5 variants in this region.

Stable chest pain is evaluated by coronary CT angiography or functional stress imaging according to pretest probability; echocardiography is first-line for new heart failure; and routine preoperative cardiac imaging is discouraged in the absence of symptoms or high-risk surgical context.

Stable Chest Pain

Region: Cardiac · ICD-10: [I20](#) , [I25](#)

Chronic stable angina and outpatient coronary artery disease evaluation. Across 2 variants, AIIE's preferred first-line studies CTA, NM.

SCENARIO DEFAULT Ccta Pathway

ccta-pathway

Applies when: age 18–65 y.

MODALITY	AIIE	RADIATION	RATIONALE
CTA ★ CT angiography	8	Moderate	Coronary CTA is appropriate for stable chest pain in low-intermediate pretest probability patients.
NM Nuclear medicine	7	Moderate	Nuclear stress imaging is an alternative when CCTA is contraindicated or calcium score is very high.
CT CT, non-contrast	4	Moderate	Non-coronary chest CT does not assess coronary stenosis in stable angina.
US Ultrasound	3	None	Echocardiography evaluates function but does not replace anatomic coronary assessment.
MRI MRI, non-contrast	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this cardiac presentation; preferred modalities address the clinical question.

MODALITY	AIIE	RADIATION	RATIONALE
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this cardiac presentation; preferred modalities address the clinical question.

Evidence anchor: [cardiac-stable-chest-pain-ccta-pathway](#)

Stress Imaging Pathway

stress-imaging-pathway

Applies when: red flags: age > 50.

MODALITY	AIIE	RADIATION	RATIONALE
NM ★ Nuclear medicine	8	Moderate	Stress myocardial perfusion imaging is appropriate for stable chest pain with intermediate pretest probability.
CTA CT angiography	7	Moderate	CCTA remains appropriate in selected patients but stress imaging is common when functional ischemia is the question.
US Ultrasound	6	None	Stress echocardiography is an alternative functional test for stable chest pain.
MRI MRI, non-contrast	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this cardiac presentation; preferred modalities address the clinical question.

Evidence anchor: [cardiac-stable-chest-pain-stress-imaging-pathway](#)

New Heart Failure

Region: Cardiac · ICD-10: [I50](#)

Newly diagnosed or decompensated heart failure imaging workup. Across 1 variant, AIIE's preferred first-line study US.

SCENARIO DEFAULT Initial Evaluation

initial-evaluation

Applies when: applies as the scenario baseline (no additional discriminators).

MODALITY	AIIE	RADIATION	RATIONALE
US ★ Ultrasound	9	None	Echocardiography is first-line for new heart failure to assess ejection fraction and valvular disease.
MRI MRI, non-contrast	7	None	Cardiac MRI characterizes cardiomyopathy and viability when echo is inconclusive.
XR Radiography (plain film)	6	Minimal	Chest radiograph assesses pulmonary edema but does not define cardiomyopathy etiology.
NM Nuclear medicine	5	Moderate	Nuclear ventriculography is secondary to echocardiography for initial HF evaluation.
MRI (C+) MRI with contrast	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this cardiac presentation; preferred modalities address the clinical question.

Evidence anchor: [cardiac-new-heart-failure-initial-evaluation](#)

Preoperative Cardiac Clearance

Region: Cardiac · ICD-10: [Z01.810](#) , [Z01.818](#)

Routine preoperative cardiac imaging without specific indication. Across 2 variants, AIIE's preferred first-line studies NM, US. Imaging is explicitly discouraged in 1 variant (low-value-care guard).

SCENARIO DEFAULT

IMAGING NOT INDICATED

Routine No Indication

routine-no-indication

Applies when: applies as the scenario baseline (no additional discriminators).

MODALITY	AIIE	RADIATION	RATIONALE
US Ultrasound	2	None	Echocardiography is reserved for patients with murmur, heart failure, or concerning history — not routine clearance.
MRI MRI, non-contrast	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Coronary CTA is not indicated for routine preoperative clearance without cardiac symptoms.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
NM ★ Nuclear medicine	1	Moderate	Routine preoperative cardiac imaging is usually not appropriate without symptoms or high-risk surgery criteria.
PET-CT PET-CT	1	High	Not indicated for this cardiac presentation; preferred modalities address the clinical question.

Evidence anchor: [cardiac-pre-op-clearance-routine-no-indication](#)

High Risk Surgery Symptoms

high-risk-surgery-symptoms

Applies when: red flags: progressive symptoms, age > 50.

MODALITY	AIIE	RADIATION	RATIONALE
US ★ Ultrasound	8	None	Echocardiography is appropriate before high-risk surgery when heart failure or valvular disease is suspected.
NM Nuclear medicine	7	Moderate	Stress imaging may be indicated for high-risk surgery with poor functional capacity.
MRI MRI, non-contrast	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this cardiac presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this cardiac presentation; preferred modalities address the clinical question.

Evidence anchor: [cardiac-pre-op-clearance-high-risk-surgery-symptoms](#)

II.8 Abdomen

6 scenarios · 11 variants in this region.

Abdominal pain is routed by quadrant, age, and pregnancy. Suspected appendicitis is ultrasound-first in children and pregnancy and CT-first in non-pregnant adults; right-upper-quadrant pain is ultrasound-first for biliary disease; obstruction and diverticulitis are CT; and blunt trauma is triaged by haemodynamic stability between FAST ultrasound and CT.

RLQ Pain / Suspected Appendicitis

Region: Abdomen · ICD-10: [K35](#), [R10.3](#)

Right lower quadrant pain evaluation for appendicitis across age groups. Across 3 variants, AIIE's preferred first-line studies US, CT.

Pregnant US MRI

pregnant-us-mri

Applies when: pregnancy.

MODALITY	AIIE	RADIATION	RATIONALE
US ★ Ultrasound	9	None	Ultrasound is first-line for pregnant patients with suspected appendicitis.
MRI MRI, non-contrast	8	None	MRI without contrast is appropriate when ultrasound is inconclusive in pregnancy.
CT CT, non-contrast	4	Moderate	CT is generally avoided in pregnancy unless alternative imaging is non-diagnostic. Contrast: Fetal radiation risk — use only when necessary.
MRI (C+) MRI with contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.

MODALITY	AIIE	RADIATION	RATIONALE
NM Nuclear medicine	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this abdominal presentation; preferred modalities address the clinical question.

Evidence anchor: [abdomen-rlq-appendicitis-pregnant-us-mri](#)

SCENARIO DEFAULT Adult CT

adult-ct

Applies when: age ≥ 18 y.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	9	Moderate	CT abdomen/pelvis with IV contrast is first-line for adult suspected appendicitis.
CT (C+) CT with IV contrast	9	Moderate	Contrast-enhanced CT is the standard for adult appendicitis diagnosis.
US Ultrasound	5	None	Ultrasound is less sensitive than CT in non-pregnant adults for appendicitis.
MRI MRI, non-contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this abdominal presentation; preferred modalities address the clinical question.

Evidence anchor: [abdomen-rlq-appendicitis-adult-ct](#)

Pediatric US First

pediatric-us-first

Applies when: red flags: age < 18; age ≤ 18 y.

MODALITY	AIIE	RADIATION	RATIONALE
US ★ Ultrasound	9	None	Ultrasound is first-line for pediatric suspected appendicitis to minimize radiation.
MRI MRI, non-contrast	8	None	MRI is appropriate when ultrasound is inconclusive in pediatric appendicitis.
CT CT, non-contrast	6	Moderate	CT is reserved when US and MRI are unavailable or non-diagnostic in children.
MRI (C+) MRI with contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this abdominal presentation; preferred modalities address the clinical question.

Evidence anchor: [abdomen-rlq-appendicitis-pediatric-us-first](#)

RUQ Pain / Suspected Cholecystitis

Region: Abdomen · ICD-10: [K81](#) , [K80](#) , [R10.1](#)

Right upper quadrant pain evaluation for biliary disease. Across 1 variant, AIIE's preferred first-line study US.

SCENARIO DEFAULT **US First**

us-first

Applies when: applies as the scenario baseline (no additional discriminators).

MODALITY	AIIE	RADIATION	RATIONALE
US ★ Ultrasound	9	None	Right upper quadrant ultrasound is first-line for suspected cholecystitis and cholelithiasis.
MRI MRI, non-contrast	5	None	MRCP may evaluate biliary ductal disease when US is limited.
CT CT, non-contrast	4	Moderate	CT is secondary when ultrasound is inconclusive or complications are suspected.
MRI (C+) MRI with contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this abdominal presentation; preferred modalities address the clinical question.

Evidence anchor: [abdomen-ruq-cholecystitis-us-first](#)

Epigastric Pain / Pancreatitis

Region: Abdomen · ICD-10: [K85](#), [R10.1](#)

Epigastric pain with suspected acute pancreatitis or complications. Across 2 variants, AIIE's preferred first-line studies CT, CT (C+). Imaging is explicitly discouraged in 1 variant (low-value-care guard).

SCENARIO DEFAULT

IMAGING NOT INDICATED

Uncomplicated

uncomplicated

Applies when: applies as the scenario baseline (no additional discriminators).

MODALITY	AIIE	RADIATION	RATIONALE
US Ultrasound	3	None	Ultrasound may assess gallstones as etiology but is not required for uncomplicated pancreatitis.
CT ★ CT, non-contrast	2	Moderate	CT is usually not appropriate for uncomplicated acute pancreatitis diagnosed clinically and by lipase.
MRI MRI, non-contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this abdominal presentation; preferred modalities address the clinical question.

Evidence anchor: [abdomen-epigastric-pancreatitis-uncomplicated](#)

Severe Or Suspected Complication

severe-or-suspected-complication

Applies when: red flags: fever / infectious concern, progressive symptoms.

MODALITY	AIIE	RADIATION	RATIONALE
CT (C+) ★ CT with IV contrast	8	Moderate	

MODALITY	AIIE	RADIATION	RATIONALE
			Contrast-enhanced CT is appropriate for severe pancreatitis or suspected necrosis and complications.
CT CT, non-contrast	7	Moderate	CT evaluates pancreatic necrosis, fluid collections, and vascular complications.
MRI MRI, non-contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this abdominal presentation; preferred modalities address the clinical question.

Evidence anchor: [abdomen-epigastric-pancreatitis-severe-or-suspected-complication](#)

Suspected Small Bowel Obstruction

Region: Abdomen · ICD-10: [K56](#)

Bowel obstruction evaluation with CT as standard. Across 1 variant, AIIE's preferred first-line study CT.

SCENARIO DEFAULT Standard

standard

Applies when: applies as the scenario baseline (no additional discriminators).

MODALITY	AIIE	RADIATION	RATIONALE
CT ★		Moderate	

MODALITY	AIIE	RADIATION	RATIONALE
CT, non-contrast	9		CT abdomen/pelvis with contrast is first-line for suspected small bowel obstruction.
CT (C+) CT with IV contrast	9	Moderate	CT with oral/IV contrast defines obstruction level, transition point, and ischemia.
XR Radiography (plain film)	5	Minimal	Abdominal radiographs may suggest SBO but CT is definitive.
US Ultrasound	3	None	Ultrasound has limited role for adult SBO diagnosis.
MRI MRI, non-contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this abdominal presentation; preferred modalities address the clinical question.

Evidence anchor: [abdomen-suspected-sbo-standard](#)

Suspected Diverticulitis

Region: Abdomen · ICD-10: [K57](#)

Left lower quadrant pain evaluation for acute diverticulitis. Across 2 variants, AIIE's preferred first-line studies CT, CT (C+).

SCENARIO DEFAULT Uncomplicated

uncomplicated

Applies when: applies as the scenario baseline (no additional discriminators).

MODALITY	AIIE	RADIATION	RATIONALE
CT ★	8	Moderate	

MODALITY	AIIE	RADIATION	RATIONALE
CT, non-contrast			CT abdomen/pelvis with contrast is appropriate for suspected uncomplicated diverticulitis.
CT (C+) CT with IV contrast	8	Moderate	Contrast CT confirms diverticulitis and excludes abscess or perforation.
US Ultrasound	3	None	Ultrasound is not standard for diverticulitis in adults.
MRI MRI, non-contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this abdominal presentation; preferred modalities address the clinical question.

Evidence anchor: [abdomen-diverticulitis-uncomplicated](#)

Complicated

complicated

Applies when: red flags: fever / infectious concern, immunocompromised host.

MODALITY	AIIE	RADIATION	RATIONALE
CT (C+) ★ CT with IV contrast	9	Moderate	Contrast CT is indicated for complicated diverticulitis to assess abscess, perforation, and fistula.
MRI MRI, non-contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRI (C+)	1	None	

MODALITY	AIIE	RADIATION	RATIONALE
MRI with contrast			Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this abdominal presentation; preferred modalities address the clinical question.

Evidence anchor: [abdomen-diverticulitis-complicated](#)

Blunt Abdominal Trauma

Region: Abdomen · ICD-10: [S36](#), [S37](#)

Traumatic abdominal injury stratified by hemodynamic stability. Across 2 variants, AIIE's preferred first-line studies CT, US.

SCENARIO DEFAULT Hemodynamically Stable

hemodynamically-stable

Applies when: red flags: recent trauma; acute trauma.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	9	Moderate	CT abdomen/pelvis with IV contrast is indicated for stable blunt abdominal trauma with concerning mechanism.
CT (C+) CT with IV contrast	9	Moderate	Contrast CT grades solid organ injury and active bleeding in stable trauma.
US		None	

MODALITY	AIIE	RADIATION	RATIONALE
Ultrasound	7		FAST ultrasound screens for free fluid but CT defines injury grade when stable.
US Doppler Ultrasound, Doppler/duplex	6	None	Doppler may assess vascular injury but CT is primary in stable BAT.
MRI MRI, non-contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this abdominal presentation; preferred modalities address the clinical question.

Evidence anchor: [abdomen-blunt-abdominal-trauma-hemodynamically-stable](#)

Hemodynamically Unstable

hemodynamically-unstable

Applies when: red flags: recent trauma, sudden / hyperacute onset; acute trauma.

MODALITY	AIIE	RADIATION	RATIONALE
US ★ Ultrasound	9	None	FAST exam is first-line in unstable blunt trauma before OR — expedite.
CT CT, non-contrast	5	Moderate	CT is deferred in unstable patients who proceed directly to operative intervention.
MRI MRI, non-contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
MRA		None	

MODALITY	AIIE	RADIATION	RATIONALE
MR angiography	1		Not indicated for this abdominal presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this abdominal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this abdominal presentation; preferred modalities address the clinical question.

Evidence anchor: [abdomen-blunt-abdominal-trauma-hemodynamically-unstable](#)

II.9 Pelvis

1 scenario · 3 variants in this region.

Transvaginal ultrasound is the first-line study for female pelvic pain, with dedicated pathways for suspected ectopic pregnancy and postmenopausal bleeding. CT is de-emphasised to spare gonadal radiation unless ultrasound is non-diagnostic.

Pelvic Pain (Female)

Region: Pelvis · ICD-10: R10.2 , N94 , 000

Non-pregnant and pregnant female pelvic pain including ectopic workup. Across 3 variants, AIIE's preferred first-line study US.

SCENARIO DEFAULT US First

us-first

Applies when: applies as the scenario baseline (no additional discriminators).

MODALITY	AIIE	RADIATION	RATIONALE
US ★ Ultrasound	9	None	Transvaginal pelvic ultrasound is first-line for female pelvic pain evaluation.
MRI MRI, non-contrast	6	None	Pelvic MRI is secondary when ultrasound is inconclusive for adnexal pathology.
CT CT, non-contrast	4	Moderate	CT exposes gonadal tissue to radiation and is not first-line for routine pelvic pain.
MRI (C+) MRI with contrast	1	None	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
NM	1	Moderate	

MODALITY	AIIE	RADIATION	RATIONALE
Nuclear medicine			Not indicated for this pelvic presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this pelvic presentation; preferred modalities address the clinical question.

Evidence anchor: [pelvis-pelvic-pain-female-us-first](#)

Ectopic Workup

ectopic-workup

Applies when: red flags: sudden / hyperacute onset; pregnancy.

MODALITY	AIIE	RADIATION	RATIONALE
US ★ Ultrasound	9	None	Transvaginal ultrasound is first-line for suspected ectopic pregnancy — expedite when unstable.
MRI MRI, non-contrast	4	None	MRI is not standard for emergent ectopic pregnancy evaluation.
CT CT, non-contrast	3	Moderate	CT is avoided in early pregnancy unless ultrasound is non-diagnostic and patient is stable.
MRI (C+) MRI with contrast	1	None	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this pelvic presentation; preferred modalities address the clinical question.

Evidence anchor: [pelvis-pelvic-pain-female-ectopic-workup](#)

Post Menopausal Bleeding

post-menopausal-bleeding

Applies when: red flags: age > 50.

MODALITY	AIIE	RADIATION	RATIONALE
US ★ Ultrasound	8	None	Pelvic ultrasound evaluates endometrial thickness and adnexal masses in postmenopausal bleeding.
MRI MRI, non-contrast	7	None	MRI may characterize endometrial and cervical pathology when US is inconclusive.
MRI (C+) MRI with contrast	1	None	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this pelvic presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this pelvic presentation; preferred modalities address the clinical question.

Evidence anchor: [pelvis-pelvic-pain-female-post-menopausal-bleeding](#)

II.10 GU / Renal

3 scenarios · 7 variants in this region.

Adult renal colic is imaged with low-dose non-contrast CT, while pregnancy and young or recurrent presentations are ultrasound-first to limit cumulative radiation. Hematuria is worked up with CT urography, and suspected testicular torsion is an emergent scrotal Doppler study.

Renal Colic

Region: GU / Renal · ICD-10: [N20](#) , [R10.3](#)

Acute flank pain evaluation for urolithiasis. Across 3 variants, AIIE's preferred first-line studies US, CT.

Pregnant US First

pregnant-us-first

Applies when: pregnancy.

MODALITY	AIIE	RADIATION	RATIONALE
US ★ Ultrasound	9	None	Ultrasound is first-line for suspected renal colic in pregnancy to avoid ionizing radiation.
MRI MRI, non-contrast	7	None	MRI without contrast is appropriate when ultrasound is inconclusive in pregnancy.
CT CT, non-contrast	4	Moderate	CT is reserved in pregnancy when US/MRI are non-diagnostic. Contrast: Fetal radiation risk.
MRI (C+) MRI with contrast	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
NM	1	Moderate	

MODALITY	AIIE	RADIATION	RATIONALE
Nuclear medicine			Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.

Evidence anchor: [gu-renal-renal-colic-pregnant-us-first](#)

SCENARIO DEFAULT Standard Low Dose CT

standard-low-dose-ct

Applies when: age ≥ 18 y.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	8	Moderate	Low-dose non-contrast CT is appropriate for adult renal colic with high sensitivity for stones.
US Ultrasound	5	None	Ultrasound may detect hydronephrosis but is less sensitive than CT for ureteral stones in adults.
XR Radiography (plain film)	3	Minimal	Abdominal radiographs detect only radiopaque stones and miss many ureteral calculi.
MRI MRI, non-contrast	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.

Evidence anchor: [gu-renal-renal-colic-standard-low-dose-ct](#)**Young Recurrent US**

young-recurrent-us

Applies when: red flags: age < 18.

MODALITY	AIIE	RADIATION	RATIONALE
US ★ Ultrasound	8	None	Ultrasound is preferred for young patients with recurrent renal colic to limit cumulative radiation.
CT CT, non-contrast	6	Moderate	CT may be used when ultrasound is non-diagnostic in recurrent pediatric colic.
MRI MRI, non-contrast	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.

Evidence anchor: [gu-renal-renal-colic-young-recurrent-us](#)**Hematuria**Region: GU / Renal · ICD-10: [R31](#)

Microscopic and gross hematuria urologic workup. Across 2 variants, AIIE's preferred first-line study CT.

SCENARIO DEFAULT Microscopic

microscopic

Applies when: applies as the scenario baseline (no additional discriminators).

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	7	Moderate	CT urography or renal CT is appropriate for asymptomatic microscopic hematuria after urologic evaluation.
CT (C+) CT with IV contrast	7	Moderate	Contrast CT urography evaluates upper tract in microscopic hematuria workup.
US Ultrasound	6	None	Renal ultrasound may be initial in younger patients with low malignancy risk.
MRI MRI, non-contrast	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.

Evidence anchor: [gu-renal-hematuria-microscopic](#)

Gross

gross

Applies when: red flags: sudden / hyperacute onset.

MODALITY	AIIE	RADIATION	RATIONALE
CT ★ CT, non-contrast	8	Moderate	

MODALITY	AIIE	RADIATION	RATIONALE
			CT urography is indicated for gross hematuria to evaluate urothelial and renal pathology.
CT (C+) CT with IV contrast	8	Moderate	Contrast-enhanced CT urography is standard for gross hematuria evaluation.
US Ultrasound	5	None	Ultrasound alone may miss urothelial malignancy in gross hematuria.
MRI MRI, non-contrast	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.

Evidence anchor: [gu-renal-hematuria-gross](#)

Scrotal Pain

Region: GU / Renal · ICD-10: [N50](#) , [N45](#)

Acute scrotal pain including testicular torsion. Across 2 variants, AIIE's preferred first-line study US Doppler.

SCENARIO DEFAULT Standard Doppler

standard-doppler

Applies when: applies as the scenario baseline (no additional discriminators).

MODALITY	AIIE	RADIATION	RATIONALE
US Doppler ★		None	Scrotal ultrasound with Doppler is first-line for acute scrotal pain.

MODALITY	AIIE	RADIATION	RATIONALE
Ultrasound, Doppler/duplex	9		
US Ultrasound	7	None	Grayscale ultrasound alone is insufficient without Doppler for torsion assessment.
MRI MRI, non-contrast	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.

Evidence anchor: [gu-renal-scrotal-pain-standard-doppler](#)

Suspected torsion

suspected-torsion

Applies when: red flags: sudden / hyperacute onset.

MODALITY	AIIE	RADIATION	RATIONALE
US Doppler ★ Ultrasound, Doppler/duplex	9	None	Emergent scrotal Doppler ultrasound is required for suspected testicular torsion — expedite to OR if positive.
MRI MRI, non-contrast	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	

MODALITY	AIIE	RADIATION	RATIONALE
			Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this GU/renal presentation; preferred modalities address the clinical question.

Evidence anchor: [gu-renal-scrotal-pain-suspected-torsion](#)

II.11 Upper Extremity MSK

1 scenario · 3 variants in this region.

Shoulder pain is radiograph-first after acute trauma to exclude dislocation and fracture; chronic rotator-cuff symptoms after failed conservative care are evaluated by ultrasound or MRI as cost- and access-appropriate alternatives.

Shoulder Pain

Region: Upper Extremity MSK · ICD-10: [M75](#) , [S43](#)

Shoulder pain including rotator cuff and acute trauma pathways. Across 3 variants, AIIE's preferred first-line studies MRI, US, XR.

SCENARIO DEFAULT Chronic Rotator Cuff MRI

chronic-rotator-cuff-mri

Applies when: duration ≥ 42 d (6+ wk); after conservative care.

MODALITY	AIIE	RADIATION	RATIONALE
MRI ★ MRI, non-contrast	8	None	MRI shoulder is appropriate for chronic rotator cuff symptoms after failed conservative care.
US Ultrasound	7	None	Shoulder ultrasound is an acceptable alternative for rotator cuff tear assessment.
XR Radiography (plain film)	5	Minimal	Radiographs evaluate arthritis and calcific tendinopathy but not cuff tear detail.
MRI (C+) MRI with contrast	1	None	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.

MODALITY	AIIE	RADIATION	RATIONALE
PET-CT PET-CT	1	High	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.

Evidence anchor: [msk-upper-shoulder-pain-chronic-rotator-cuff-mri](#)

Chronic Rotator Cuff US

chronic-rotator-cuff-us

Applies when: duration ≥ 42 d (6+ wk).

MODALITY	AIIE	RADIATION	RATIONALE
MRI MRI, non-contrast	8	None	MRI remains gold standard when ultrasound operator expertise is limited.
US ★ Ultrasound	8	None	Shoulder ultrasound is appropriate for chronic rotator cuff evaluation as a cost-effective alternative to MRI.
MRI (C+) MRI with contrast	1	None	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.

Evidence anchor: [msk-upper-shoulder-pain-chronic-rotator-cuff-us](#)

Acute Trauma Xr First

acute-trauma-xr-first

Applies when: red flags: recent trauma; acute trauma.

MODALITY	AIIE	RADIATION	RATIONALE
XR ★ Radiography (plain film)	9	Minimal	Shoulder radiographs are first-line for acute shoulder trauma to assess dislocation and fracture.
CT CT, non-contrast	6	Moderate	CT may follow radiographs for complex fracture characterization.
MRI MRI, non-contrast	5	None	MRI is not first-line before radiographs in acute shoulder trauma.
MRI (C+) MRI with contrast	1	None	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this upper-extremity MSK presentation; preferred modalities address the clinical question.

Evidence anchor: [msk-upper-shoulder-pain-acute-trauma-xr-first](#)

II.12 Lower Extremity MSK

3 scenarios · 8 variants in this region.

Lower-extremity MSK imaging is decision-rule driven. The Ottawa knee and ankle rules gate radiographs after trauma — and AIIE explicitly discourages films when those rules are negative. MRI follows non-diagnostic radiographs for suspected internal derangement or occult hip fracture, and weight-bearing films assess osteoarthritis only when they will change management.

Knee Pain

Region: Lower Extremity MSK · ICD-10: [M17](#) , [M23](#) , [S80](#)

Knee pain triage including trauma decision rules and chronic OA. Across 4 variants, AIIE's preferred first-line studies XR, MRI. Imaging is explicitly discouraged in 2 variants (low-value-care guard).

Acute Trauma Xr Criteria

acute-trauma-xr-criteria

Applies when: red flags: recent trauma; acute trauma.

MODALITY	AIIE	RADIATION	RATIONALE
XR ★ Radiography (plain film)	8	Minimal	Knee radiographs are indicated when Ottawa knee rules criteria are met after acute trauma.
MRI MRI, non-contrast	5	None	MRI is not first-line before radiographs in acute knee trauma.
MRI (C+) MRI with contrast	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	

MODALITY	AIIE	RADIATION	RATIONALE
			Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.

Evidence anchor: [msk-lower-knee-pain-acute-trauma-xr-criteria](#)

IMAGING NOT INDICATED Acute Trauma Low Risk

acute-trauma-low-risk

Applies when: red flags: recent trauma; acute trauma; after conservative care.

MODALITY	AIIE	RADIATION	RATIONALE
MRI MRI, non-contrast	2	None	MRI is not indicated for low-risk acute knee injury meeting decision rule criteria.
XR ★ Radiography (plain film)	2	Minimal	Radiographs are usually not appropriate when Ottawa knee rules are negative.
MRI (C+) MRI with contrast	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
PET-CT		High	

MODALITY	AIIE	RADIATION	RATIONALE
PET-CT	1		Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.

Evidence anchor: [msk-lower-knee-pain-acute-trauma-low-risk](#)

Meniscal Ligament Post Xr

meniscal-ligament-post-xr

Applies when: prior imaging present; after conservative care.

MODALITY	AIIE	RADIATION	RATIONALE
MRI ★ MRI, non-contrast	8	None	MRI knee is appropriate for suspected meniscal or ligament injury after non-diagnostic radiographs.
XR Radiography (plain film)	3	Minimal	Repeat radiographs add little when internal derangement is suspected post-initial XR.
MRI (C+) MRI with contrast	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.

Evidence anchor: [msk-lower-knee-pain-meniscal-ligament-post-xr](#)

SCENARIO DEFAULT

IMAGING NOT INDICATED

Chronic OA

chronic-oa

Applies when: duration ≥ 90 d (12+ wk).

MODALITY	AIIE	RADIATION	RATIONALE
XR ★ Radiography (plain film)	6	Minimal	Weight-bearing knee radiographs may assess osteoarthritis severity when management changes.
MRI MRI, non-contrast	2	None	MRI is usually not appropriate for uncomplicated chronic knee osteoarthritis without mechanical symptoms.
MRI (C+) MRI with contrast	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.

Evidence anchor: [msk-lower-knee-pain-chronic-oa](#)

Hip Pain

Region: Lower Extremity MSK · ICD-10: [M16](#) , [S72](#)

Hip pain including occult fracture evaluation. Across 2 variants, AIIE's preferred first-line studies MRI, XR.

Occult Fracture Negative Xr

occult-fracture-negative-xr

Applies when: red flags: osteoporosis / fragility, recent trauma; prior imaging present.

MODALITY	AIIE	RADIATION	RATIONALE
MRI ★ MRI, non-contrast	9	None	MRI hip is indicated for suspected occult fracture when radiographs are negative.
CT CT, non-contrast	6	Moderate	CT may detect fracture but MRI is more sensitive for marrow edema and occult injury.
XR Radiography (plain film)	3	Minimal	Repeat radiographs have low yield after negative initial films with high suspicion.
MRI (C+) MRI with contrast	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.

Evidence anchor: [msk-lower-hip-pain-occult-fracture-negative-xr](#)

SCENARIO DEFAULT **Acute Trauma**

acute-trauma

Applies when: red flags: recent trauma; acute trauma.

MODALITY	AIIE	RADIATION	RATIONALE
XR ★ Radiography (plain film)	9	Minimal	Hip radiographs are first-line for acute hip trauma and suspected fracture.
CT CT, non-contrast	7	Moderate	CT may define fracture comminution when surgical planning requires detail.
MRI MRI, non-contrast	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.

Evidence anchor: [msk-lower-hip-pain-acute-trauma](#)

Ankle Pain

Region: Lower Extremity MSK · ICD-10: [S82](#) , [S93](#)

Ankle injury triage using Ottawa ankle rules. Across 2 variants, AIIE's preferred first-line study XR. Imaging is explicitly discouraged in 1 variant (low-value-care guard).

SCENARIO DEFAULT Trauma Meets Criteria

trauma-meets-criteria

Applies when: red flags: recent trauma; acute trauma.

MODALITY	AIIE	RADIATION	RATIONALE
XR ★ Radiography (plain film)	8	Minimal	Ankle radiographs are indicated when Ottawa ankle rules criteria are met.
CT CT, non-contrast	4	Moderate	CT is not first-line for routine ankle trauma meeting XR indications.
MRI MRI, non-contrast	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.

Evidence anchor: [msk-lower-ankle-pain-trauma-meets-criteria](#)
IMAGING NOT INDICATED Trauma Low Risk

trauma-low-risk

Applies when: red flags: recent trauma; acute trauma; after conservative care.

MODALITY	AIIE	RADIATION	RATIONALE
XR ★ Radiography (plain film)	2	Minimal	Radiographs are usually not appropriate when Ottawa ankle rules are negative.

MODALITY	AIIE	RADIATION	RATIONALE
MRI MRI, non-contrast	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this lower-extremity MSK presentation; preferred modalities address the clinical question.

Evidence anchor: [msk-lower-ankle-pain-trauma-low-risk](#)

II.13 Vascular

3 scenarios · 5 variants in this region.

Duplex ultrasound is the vascular workhorse across suspected DVT, carotid stenosis, and abdominal aortic aneurysm screening and surveillance. CT and MR angiography are reserved for confirmation, inconclusive ultrasound, or pre-procedural planning.

Suspected DVT

Region: Vascular · ICD-10: [I82](#), [R60](#)

Lower extremity deep venous thrombosis evaluation. Across 1 variant, AIIE's preferred first-line study US Doppler.

SCENARIO DEFAULT Standard Doppler

standard-doppler

Applies when: applies as the scenario baseline (no additional discriminators).

MODALITY	AIIE	RADIATION	RATIONALE
US Doppler ★ Ultrasound, Doppler/duplex	9	None	Lower extremity venous duplex ultrasound is first-line for suspected DVT.
MRA MR angiography	4	None	MRV is secondary to duplex ultrasound for routine DVT diagnosis.
CTA CT angiography	4	Moderate	CT venography is reserved when ultrasound is inconclusive or central clot is suspected.
MRI MRI, non-contrast	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this vascular presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this vascular presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this vascular presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this vascular presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this vascular presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this vascular presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	

MODALITY	AIIE	RADIATION	RATIONALE
			Not indicated for this vascular presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this vascular presentation; preferred modalities address the clinical question.

Evidence anchor: [vascular-suspected-dvt-standard-doppler](#)

Carotid Stenosis Workup

Region: Vascular · ICD-10: [I65](#), [I63](#)

Symptomatic and asymptomatic carotid artery stenosis evaluation. Across 2 variants, AIIE's preferred first-line study US Doppler.

SCENARIO DEFAULT Symptomatic

symptomatic

Applies when: red flags: sudden / hyperacute onset.

MODALITY	AIIE	RADIATION	RATIONALE
US Doppler ★ Ultrasound, Doppler/duplex	9	None	Carotid duplex ultrasound is first-line for symptomatic carotid stenosis workup.
MRA MR angiography	8	None	MRA neck is an alternative when CTA contrast is contraindicated.
CTA CT angiography	8	Moderate	CTA neck confirms stenosis degree when ultrasound is limited or preoperative planning requires detail.
MRI MRI, non-contrast	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this vascular presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this vascular presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this vascular presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this vascular presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this vascular presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this vascular presentation; preferred modalities address the clinical question.
NM		Moderate	

MODALITY	AIIE	RADIATION	RATIONALE
Nuclear medicine	1		Not indicated for this vascular presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this vascular presentation; preferred modalities address the clinical question.

Evidence anchor: [vascular-carotid-stenosis-symptomatic](#)

Asymptomatic Screening

asymptomatic-screening

Applies when: applies as the scenario baseline (no additional discriminators).

MODALITY	AIIE	RADIATION	RATIONALE
US Doppler ★ Ultrasound, Doppler/duplex	7	None	Carotid duplex is appropriate for asymptomatic bruit or high-risk screening when criteria are met.
CTA CT angiography	5	Moderate	CTA is secondary to duplex for asymptomatic carotid screening.
MRI MRI, non-contrast	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this vascular presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this vascular presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this vascular presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this vascular presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this vascular presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this vascular presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this vascular presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this vascular presentation; preferred modalities address the clinical question.

Evidence anchor: [vascular-carotid-stenosis-asymptomatic-screening](#)

AAA Screening / Surveillance

Region: Vascular · ICD-10: [I71](#) , [Z13.6](#)

Abdominal aortic aneurysm screening and surveillance imaging. Across 2 variants, AIIE's preferred first-line study US.

SCENARIO DEFAULT Screening Eligible

screening-eligible

Applies when: age 65–75 y.

MODALITY	AIIE	RADIATION	RATIONALE
US ★ Ultrasound	9	None	One-time abdominal ultrasound is appropriate AAA screening for eligible male ever-smokers age 65–75.
CT CT, non-contrast	5	Moderate	CT is not first-line for AAA screening due to radiation; used when US is inconclusive.
CTA CT angiography	4	Moderate	CTA is for aneurysm characterization pre-repair, not population screening.
MRI MRI, non-contrast	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this vascular presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this vascular presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this vascular presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this vascular presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this vascular presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this vascular presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this vascular presentation; preferred modalities address the clinical question.

Evidence anchor: [vascular-aaa-screening-screening-eligible](#)

Surveillance Known AAA

surveillance-known-aaa

Applies when: prior imaging present.

MODALITY	AIIE	RADIATION	RATIONALE
US ★ Ultrasound	9	None	Ultrasound surveillance is standard for known small AAA per size-based intervals.
CTA CT angiography	7	Moderate	CTA is appropriate pre-endovascular or open repair planning for enlarging AAA.
CT CT, non-contrast	6	Moderate	CT may define aneurysm morphology when repair planning requires detail.
MRI MRI, non-contrast	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this vascular presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this vascular presentation; preferred modalities address the clinical question.
Mammo Mammography	1	Minimal	Not indicated for this vascular presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this vascular presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this vascular presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this vascular presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this vascular presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this vascular presentation; preferred modalities address the clinical question.

Evidence anchor: [vascular-aaa-screening-surveillance-known-aaa](#)

II.14 Breast

2 scenarios · 4 variants in this region.

Mammography is the screening foundation for average-risk women from age 40. A palpable lump is ultrasound-first under 30 and mammography-plus-targeted-ultrasound at 30 and older, and supplemental MRI is reserved for high-risk populations rather than average risk.

Breast Cancer Screening

Region: Breast · ICD-10: [Z12.3](#)

Population screening mammography by age and risk tier. Across 2 variants, AIIE's preferred first-line studies Mammo, MRI.

SCENARIO DEFAULT Average Risk 40 +

average-risk-40-plus

Applies when: age 40–74 y.

MODALITY	AIIE	RADIATION	RATIONALE
Mammo ★ Mammography	9	Minimal	Screening mammography is appropriate for average-risk women starting at age 40 per society guidelines.
US Ultrasound	3	None	Ultrasound is not a replacement for screening mammography in average-risk women.
MRI MRI, non-contrast	2	None	Breast MRI screening is reserved for high-risk populations, not average risk.
MRI (C+) MRI with contrast	1	None	Not indicated for this breast presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this breast presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this breast presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this breast presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this breast presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this breast presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this breast presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this breast presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this breast presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	

MODALITY	AIIE	RADIATION	RATIONALE
			Not indicated for this breast presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this breast presentation; preferred modalities address the clinical question.

Evidence anchor: [breast-breast-screening-average-risk-40-plus](#)

High Risk MRI Supplement

high-risk-mri-supplement

Applies when: red flags: prior cancer history, immunocompromised host.

MODALITY	AIIE	RADIATION	RATIONALE
Mammo Mammography	9	Minimal	Mammography remains the foundation even in high-risk screening protocols.
MRI ★ MRI, non-contrast	8	None	Supplemental breast MRI is appropriate for high-risk women per ACS guidelines.
MRI (C+) MRI with contrast	1	None	Not indicated for this breast presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this breast presentation; preferred modalities address the clinical question.
US Ultrasound	1	None	Not indicated for this breast presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this breast presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this breast presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this breast presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this breast presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this breast presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this breast presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this breast presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this breast presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this breast presentation; preferred modalities address the clinical question.

Evidence anchor: [breast-breast-screening-high-risk-mri-supplement](#)

Palpable Breast Lump

Region: Breast · ICD-10: [N63](#) , [R22](#)

Clinical breast lump evaluation stratified by age. Across 2 variants, AIIE's preferred first-line studies US, Mammo.

under 30 US First

[under-30-us-first](#)

Applies when: age ≤ 29 y.

MODALITY	AIIE	RADIATION	RATIONALE
US ★ Ultrasound	9	None	Ultrasound is first-line for palpable breast lump in women under 30.
Mammo Mammography	5	Minimal	Diagnostic mammography may supplement ultrasound in selected young patients.
MRI MRI, non-contrast	1	None	Not indicated for this breast presentation; preferred modalities address the clinical question.
MRI (C+) MRI with contrast	1	None	Not indicated for this breast presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this breast presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this breast presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this breast presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this breast presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this breast presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this breast presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this breast presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this breast presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this breast presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this breast presentation; preferred modalities address the clinical question.

Evidence anchor: [breast-palpable-lump-under-30-us-first](#)

SCENARIO DEFAULT 30 + Mammo US

30-plus-mammo-us

Applies when: age ≥ 30 y.

MODALITY	AIIE	RADIATION	RATIONALE
US Ultrasound	9	None	Targeted breast ultrasound complements mammography for palpable masses age 30+.
Mammo ★ Mammography	9	Minimal	Diagnostic mammography plus targeted ultrasound is appropriate for palpable lump age 30 and older.
MRI MRI, non-contrast	6	None	Breast MRI is secondary for problem-solving when mammo/US are inconclusive.
MRI (C+) MRI with contrast	1	None	Not indicated for this breast presentation; preferred modalities address the clinical question.
MRA MR angiography	1	None	Not indicated for this breast presentation; preferred modalities address the clinical question.
US Doppler Ultrasound, Doppler/duplex	1	None	Not indicated for this breast presentation; preferred modalities address the clinical question.
XR Radiography (plain film)	1	Minimal	Not indicated for this breast presentation; preferred modalities address the clinical question.
DEXA DEXA densitometry	1	Minimal	Not indicated for this breast presentation; preferred modalities address the clinical question.
Fluoro Fluoroscopy	1	Low	Not indicated for this breast presentation; preferred modalities address the clinical question.
CT CT, non-contrast	1	Moderate	Not indicated for this breast presentation; preferred modalities address the clinical question.
CTA CT angiography	1	Moderate	Not indicated for this breast presentation; preferred modalities address the clinical question.
CT (C+) CT with IV contrast	1	Moderate	Not indicated for this breast presentation; preferred modalities address the clinical question.
NM Nuclear medicine	1	Moderate	Not indicated for this breast presentation; preferred modalities address the clinical question.
PET-CT PET-CT	1	High	Not indicated for this breast presentation; preferred modalities address the clinical question.

Evidence anchor: [breast-palpable-lump-30-plus-mammo-us](#)

PART III

Evidence & Intellectual-Property Basis

How AIIE relates to the ACR Appropriateness Criteria® without reproducing them, why independent derivation is the durable legal position, and how to describe AIIE's performance truthfully so the product keeps both its credibility and its Non-Device CDS posture.

IN THIS PART

1. III.1 The factual position on the ACR Appropriateness Criteria®
2. III.2 Why 'relabel and claim superior' is the wrong fix
3. III.3 AIIE's actual protection: independent derivation
4. III.4 Trademark & attribution
5. III.5 Performance-claim discipline
6. III.6 Recommended notices & on-screen language
7. III.7 Open items & counsel sign-off

III.1 The factual position on the ACR Appropriateness Criteria®

The premise that the ACR Appropriateness Criteria® were "bought by a private company" is, on the evidence available as of June 2026, not correct — and getting this right changes the whole strategy.

The ACR Appropriateness Criteria® are owned, developed, and published by the **American College of Radiology**, a non-profit professional society, and are freely accessible through the ACR's public search at acsearch.acr.org. They comprise roughly 247 topics, ~1,200 variants, and ~4,000 clinical scenarios, maintained by expert panels. They are **copyrighted** — under U.S. and international copyright — and have been for their entire history. No public record indicates an acquisition of the Criteria by a private company.

WHAT IS AND ISN'T TRUE

True: the ACR text is copyrighted, so copying or transcribing it into a product is a real legal risk. **Not true:** that the Criteria were privately acquired, or that merely *aligning* to the same underlying evidence creates liability. Copyright protects the ACR's particular expression — not the medical facts and not the appropriateness concept itself.

III.2 Why 'relabel and claim superior' is the wrong fix

Taking ACR content, renaming the ratings "AIIE," and presenting the result as an original, superior engine would create more exposure than it removes — on two independent fronts.

- **Copyright.** Reproducing or closely paraphrasing the ACR's scenario text or rating tables produces a *derivative work*. Renaming it does not cure infringement; it can aggravate it by stripping attribution.
- **Misrepresentation / passing-off.** Presenting copied content as ARKA's own novel invention, and as "better than ACR," layers a false-origin and false-superiority claim on top of the copyright problem — exactly the kind of statement that draws scrutiny in clinical and investment contexts.

THE TRAP TO AVOID

The project's own engineering rule already says it plainly: "*NEVER label anything 'ACR rating' or reproduce ACR Appropriateness Criteria content.*" Relabeling ACR content as proprietary AIIE content would break that rule and remove the very protection described next.

III.3 AIIE's actual protection: independent derivation

AIIE is already built the defensible way. Its ratings are derived independently from **primary, openly citable evidence** — the methodologies, decision rules, and society guidance catalogued in Appendix D — and the engine surfaces those citations on every recommendation. The Clinical Knowledge Matrix is ARKA's own expression: its scenarios, variant discriminators, rating values, and rationales were authored by ARKA, not copied. This is what makes the work both original and defensible.

- Ratings are expressed as **AIIE scores** on ARKA's 1–9 scale, with the field name `aiieScore`; nothing is labelled an "ACR rating."
- Every factor carries a citation to a primary source the clinician can verify independently.
- Alignment to the same three-band clinical logic the field uses is permissible; it is the shared evidence and the shared concept of appropriateness, not the ACR's text.

BOTTOM LINE

Independent derivation plus transparent primary citation is the shield. It is already implemented. Keep it — do not trade it for a relabeling shortcut.

III.4 Trademark and attribution

"ARKA", "AIIE", "ARKA Imaging Intelligence Engine", and "remARKAbly precise" are marks of ARKA Health and should be used consistently. "ACR Appropriateness Criteria®" is a registered mark of the American College of Radiology and should appear only in nominative, factual references (e.g. describing alignment or contrast), always with its ® and never in a way that implies endorsement.

RECOMMENDED STANDING NOTICE

"ARKA and AIIE are independent of, and not affiliated with, sponsored by, or endorsed by the American College of Radiology. AIIE appropriateness scores are generated by ARKA's own evidence-based engine and are not ACR Appropriateness Criteria® ratings."

III.5 Performance-claim discipline

The single most valuable asset in a regulated, clinical sale is a claim you can defend. AIIE's positioning should therefore state only what has been measured and clearly label what has not.

Guidance	Language
Say	"74% three-class accuracy on our synthetic, appropriateness-aligned validation cohort; real-world AUC pending pilot data."
Say	"Transparent, non-device CDS: every recommendation shows its factors and citations so the clinician stays the decision-maker."
Don't say	"Better than ACR" / "more accurate than the ACR Appropriateness Criteria" — no head-to-head study supports it.
Don't say	"Validated" or "clinically proven" without naming the cohort; "FDA-approved" (the posture is Non-Device CDS, which is not a clearance).

HONESTY IS THE MOAT

The same restraint that keeps the claims defensible also satisfies the transparency criterion of Non-Device CDS and is, per the project's own demo guidance, the credibility advantage over black-box competitors. Under-claiming is the strategy, not a compromise.

III.6 Recommended notices and on-screen language

- Label every score an **AIIE score** with the 1–9 band; show the contributing factors and at least one citation.
- Carry the Non-Device CDS disclosure and the "not affiliated with ACR" notice in product and marketing surfaces.
- State the measured metric paired with "real-world AUC pending" wherever performance is mentioned.
- Keep the clinician as decision-maker: recommendations are advisory and overridable, never automatic.

III.7 Open items and counsel sign-off

NOT LEGAL ADVICE

This part is informational positioning guidance, not a legal opinion. The analysis should be reviewed and signed off by qualified IP and regulatory counsel before external use. The project's [SCOPE_BOUNDARY.md](#) currently records counsel sign-off as *pending*; closing that item is the recommended next step, along with a documented evidence-source register (Appendix D) and a trademark-use checklist.

APPENDICES

Reference Material

Scale legend, modality and radiation dictionary, red-flag dictionary, the primary-evidence register, region-default ratings, the CPT and ICD-10 cross-references, a modality-first appropriateness index, methodology and change control, coverage statistics, and a glossary.

IN THIS PART

1. A • AIIE 1-9 scale
2. B • Modality & radiation dictionary
3. C • Red-flag dictionary
4. D • Primary-evidence register
5. E • Region-default (Tier 3) ratings
6. F • CPT to modality/region reference
7. G • ICD-10 to scenario cross-reference
8. H • Modality-first appropriateness index
9. I • Methodology & change control
10. J • Coverage statistics
11. K • Glossary

Appendix A — The AIIE 1-9 appropriateness scale

1 - 3 Usually not appropriate	4 - 6 May be appropriate	7 - 9 Usually appropriate
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Band	Label	Clinician action
1 1-3	Usually not appropriate	Discourage; document why if overriding.
4 4-6	May be appropriate	Conditional; depends on documentation, sequence, shared decision-making.
7 7-9	Usually appropriate	Support as first-line or confirmatory study.

Appendix B — Modality & radiation dictionary

Short	Modality	Engine key	Radiation tier
XR	Radiography (plain film)	xr	1 - Minimal
CT	CT, non-contrast	ct	3 - Moderate
CTA	CT angiography	cta	3 - Moderate
CT (C+)	CT with IV contrast	ct_contrast	3 - Moderate
MRI	MRI, non-contrast	mri	0 - None
MRI (C+)	MRI with contrast	mri_contrast	0 - None
MRA	MR angiography	mra	0 - None
US	Ultrasound	us	0 - None
US Doppler	Ultrasound, Doppler/duplex	us_doppler	0 - None
NM	Nuclear medicine	nm	3 - Moderate
PET-CT	PET-CT	pet_ct	4 - High
Fluoro	Fluoroscopy	fluoro	2 - Low
Mammo	Mammography	mammo	1 - Minimal
DEXA	DEXA densitometry	dexa	1 - Minimal

Radiation tiers are relative burden indicators (0 none -> 4 high), not absolute dose values.

Appendix C — Red-flag discriminator dictionary

Key	Meaning	Effect in AIIE
cancerHistory	prior cancer history	Selects malignancy/metastasis variants; severity boost in red-flag factor.
neurologicalDeficit	focal neurologic deficit	Selects high-acuity neuro variants; severity boost.
fever	fever / infectious concern	Selects infection/abscess variants.
weightLoss	unintentional weight loss	Selects malignancy work-up variants.
trauma	recent trauma	Activates trauma variants and the trauma-severity gate.
immunocompromised	immunocompromised host	Selects infection variants; raises patient-risk signal.
ivDrugUse	IV drug use	Selects spinal/epidural infection variants.
osteoporosis	osteoporosis / fragility	Selects fragility-fracture variants.
ageOver50	age > 50	Selects new-onset-after-50 and screening variants.
ageUnder18	age < 18	Selects pediatric, radiation-minimising variants.
progressiveSymptoms	progressive symptoms	Selects progressive-disease variants.
bladderBowelDysfunction	bladder/bowel dysfunction	Selects cauda-equina / emergent variants; severity boost.
suddenOnset	sudden / hyperacute onset	Selects hyperacute variants (SAH, dissection, torsion, stroke).

Appendix D — Primary-evidence register

The evidence bodies AIIE derives from and cites. Citation indicates the source informs ARKA's independent ratings; it does not imply endorsement by the cited organisations.

Evidence source	Domain	How AIIE uses it
RAND/UCLA Appropriateness Method	Appropriateness methodology	Framing of indication strength and the 1-9 construct.
GRADE (evidence-to-decision)	Evidence grading	Transparent factor contributions and rationale framing.
Choosing Wisely	Low-value care	Redundant-imaging penalties; imaging-not-indicated variants.
USPSTF	Preventive screening	Lung LDCT and AAA screening eligibility windows.
ACEP clinical policies	Emergency medicine	Red-flag escalation for time-critical diagnoses.
PECARN	Pediatric head injury	Low-risk pediatric head-trauma variants (observation over CT).
Ottawa Knee & Ankle Rules	MSK decision rules	Trauma radiograph indication vs low-risk no-imaging variants.
Canadian C-Spine Rule / NEXUS	Cervical trauma	CT-vs-clear decision for blunt cervical trauma.
Wells / pretest probability + D-dimer	Venous thromboembolism	PE pathway variants (low/intermediate/high).
BEIR VII / ICRP	Radiation protection	Radiation-burden counterweight and dose-aware preferences.
AAST; ISS / AIS / RTS / REMS / GCS	Trauma severity	Trauma-severity gate signal from structured scores.
ACS / society mammography guidance	Breast	Screening and high-risk supplemental-MRI variants.
ACR TI-RADS	Thyroid	Thyroid-nodule ultrasound-first characterisation.
ASHA practice guidance	Swallowing	VFSS/FEES/bedside swallow triage.
NIHSS-aligned lexicon	Stroke	STAT-gate stroke recognition.
ICD-10-CM / CPT; curated payer policy	Coding & necessity	Normalization and the MNAI alignment index.

Appendix E — Region-default (Tier 3) ratings

When region and modality are known but no scenario matches, AIIE returns a conservative mid-scale rating (workhorse modalities 5 = indeterminate; others 4) with a documentation-improvement prompt - never an unearned endorsement.

Region	Default ratings
Head / Brain	CT 5 , MRI 5 , CTA 4 , MRA 4 · all others 4
Head, Face & Neck	CT 5 , US 5 , MRI 4 · all others 4
Cervical Spine	MRI 5 , CT 4 , XR 4 · all others 4
Thoracic Spine	MRI 5 , CT 4 , XR 4 · all others 4
Lumbar Spine	MRI 5 , CT 5 , XR 4 · all others 4
Chest	XR 5 , CT 5 , CTA 4 · all others 4
Cardiac	US 5 , CTA 4 , NM 4 · all others 4
Abdomen	CT 5 , CT (C+) 5 , US 5 · all others 4
Pelvis	US 5 , MRI 4 , CT 4 · all others 4
GU / Renal	CT 5 , US 5 , US Doppler 5 · all others 4
Upper Extremity MSK	XR 5 , MRI 4 , US 4 · all others 4
Lower Extremity MSK	XR 5 , MRI 4 · all others 4
Vascular	US Doppler 5 , CTA 4 , MRA 4 · all others 4
Breast	Mammo 5 , US 5 · all others 4

Appendix F — CPT to modality / region reference

The imaging CPT subranges the normalizer resolves directly to a modality and, where anatomically specific, a body region. Reproduced from the engine source for audit.

CPT range	Modality	Region
70010-70390	XR Radiography (plain film)	Head, Face & Neck
70450-70470	CT CT, non-contrast	Head / Brain
70480-70492	CT CT, non-contrast	Head, Face & Neck
70496-70498	CTA CT angiography	Head / Brain
70540-70543	MRI MRI, non-contrast	Head, Face & Neck
70544-70549	MRA MR angiography	Head / Brain
70551-70559	MRI MRI, non-contrast	Head / Brain
71045-71048	XR Radiography (plain film)	Chest
71250-71270	CT CT, non-contrast	Chest
71271-71275	CTA CT angiography	Chest
72020	XR Radiography (plain film)	—
72040-72052	XR Radiography (plain film)	Cervical Spine
72070-72074	XR Radiography (plain film)	Thoracic Spine
72100-72120	XR Radiography (plain film)	Lumbar Spine
72125-72127	CT CT, non-contrast	Cervical Spine
72128-72130	CT CT, non-contrast	Thoracic Spine
72131-72133	CT CT, non-contrast	Lumbar Spine
72141-72142	MRI MRI, non-contrast	Cervical Spine
72146-72147	MRI MRI, non-contrast	Thoracic Spine
72148-72149	MRI MRI, non-contrast	Lumbar Spine
72156	MRI MRI, non-contrast	Cervical Spine
72157	MRI MRI, non-contrast	Thoracic Spine
72158	MRI MRI, non-contrast	Lumbar Spine
72191	CTA CT angiography	Pelvis
72192-72194	CT CT, non-contrast	Pelvis
72195-72197	MRI MRI, non-contrast	Pelvis
72198	MRA MR angiography	Pelvis

CPT range	Modality	Region
73000-73140	XR Radiography (plain film)	Upper Extremity MSK
73200-73202	CT CT, non-contrast	Upper Extremity MSK
73206	CTA CT angiography	Upper Extremity MSK
73218-73223	MRI MRI, non-contrast	Upper Extremity MSK
73225	MRA MR angiography	Upper Extremity MSK
73501-73660	XR Radiography (plain film)	Lower Extremity MSK
73700-73702	CT CT, non-contrast	Lower Extremity MSK
73706	CTA CT angiography	Lower Extremity MSK
73718-73723	MRI MRI, non-contrast	Lower Extremity MSK
73725	MRA MR angiography	Lower Extremity MSK
74018-74022	XR Radiography (plain film)	Abdomen
74150-74170	CT CT, non-contrast	Abdomen
74174-74175	CTA CT angiography	Abdomen
74176-74178	CT CT, non-contrast	Abdomen
74181-74183	MRI MRI, non-contrast	Abdomen
74185	MRA MR angiography	Abdomen
75557-75565	MRI MRI, non-contrast	Cardiac
75571-75573	CT CT, non-contrast	Cardiac
75574	CTA CT angiography	Cardiac
76000-76120	Fluoro Fluoroscopy	—
76506-76999	US Ultrasound	—
77046-77049	MRI MRI, non-contrast	Breast
77065-77067	Mammo Mammography	Breast
77080-77086	DEXA DEXA densitometry	—
78012-78799	NM Nuclear medicine	—
78811-78816	PET-CT PET-CT	Whole Body
93880-93882	US Doppler Ultrasound, Doppler/duplex	Vascular

CPT range	Modality	Region
93925-93931	US Doppler Ultrasound, Doppler/duplex	Vascular
93970-93971	US Doppler Ultrasound, Doppler/duplex	Vascular
93975-93979	US Doppler Ultrasound, Doppler/duplex	Vascular

Appendix G — ICD-10 to scenario cross-reference

Each ICD-10-CM prefix the matrix recognises and the scenario(s) it supports during candidate ranking.

ICD-10 prefix	Scenario(s)
D34	Thyroid Nodule
E04	Thyroid Nodule
G40	Seizure
G44	Acute Headache
G45	Suspected Stroke
I20	Acute Chest Pain; Stable Chest Pain
I25	Stable Chest Pain
I26	Acute Chest Pain
I50	New Heart Failure
I63	Carotid Stenosis Workup; Suspected Stroke
I64	Suspected Stroke
I65	Carotid Stenosis Workup
I71	AAA Screening / Surveillance; Acute Chest Pain
I82	Suspected DVT
J01	Sinusitis
J32	Sinusitis
K35	RLQ Pain / Suspected Appendicitis
K56	Suspected Small Bowel Obstruction
K57	Suspected Diverticulitis
K80	RUQ Pain / Suspected Cholecystitis
K81	RUQ Pain / Suspected Cholecystitis
K85	Epigastric Pain / Pancreatitis
M16	Hip Pain
M17	Knee Pain
M23	Knee Pain
M48	Low Back Pain
M50	Neck Pain

ICD-10 prefix	Scenario(s)
M51	Low Back Pain; Thoracic Back Pain
M54	Low Back Pain
M54 . 2	Neck Pain
M54 . 6	Thoracic Back Pain
M75	Shoulder Pain
N20	Renal Colic
N45	Scrotal Pain
N50	Scrotal Pain
N63	Palpable Breast Lump
N94	Pelvic Pain (Female)
000	Pelvic Pain (Female)
R04 . 2	Hemoptysis
R05	Chronic Cough
R07	Acute Chest Pain
R10 . 1	Epigastric Pain / Pancreatitis; RUQ Pain / Suspected Cholecystitis
R10 . 2	Pelvic Pain (Female)
R10 . 3	RLQ Pain / Suspected Appendicitis; Renal Colic
R22	Neck Mass (Adult); Palpable Breast Lump
R31	Hematuria
R51	Acute Headache
R56	Seizure
R59	Neck Mass (Adult)
R60	Suspected DVT
S06	Head Trauma
S09	Head Trauma
S13	Neck Pain
S22	Thoracic Back Pain

ICD-10 prefix	Scenario(s)
S36	Blunt Abdominal Trauma
S37	Blunt Abdominal Trauma
S43	Shoulder Pain
S72	Hip Pain
S80	Knee Pain
S82	Ankle Pain
S93	Ankle Pain
Z01.810	Preoperative Cardiac Clearance
Z01.818	Preoperative Cardiac Clearance
Z12.2	Lung Cancer Screening
Z12.3	Breast Cancer Screening
Z13.6	AAA Screening / Surveillance
Z87.891	Lung Cancer Screening

Appendix H — Modality-first appropriateness index

For each modality, the clinical situations in which AIIE rates it **usually appropriate** (7-9). A different lens on the same matrix: read it when you have a modality and want its strongest indications. Modalities with no usually-appropriate variant in the current knowledge base are noted as such.

XR — RADIOGRAPHY (PLAIN FILM)

Chronic Cough — Standard Workup (**7** ★) · Shoulder Pain — Acute Trauma Xr First (**9** ★) · Knee Pain — Acute Trauma Xr Criteria (**8** ★) · Hip Pain — Acute Trauma (**9** ★) · Ankle Pain — Trauma Meets Criteria (**8** ★)

CT — CT, NON-CONTRAST

Acute Headache — Thunderclap (**9** ★) · Acute Headache — Neuro Deficit (**8**) · Acute Headache — New over 50 (**7**) · Acute Headache — Post Trauma GCS under 15 (**9** ★) · Suspected Stroke — Acute under 4 5h (**9** ★) · Suspected Stroke — Wake Up Stroke (**8**) · Seizure — First Seizure (**7**) · Head Trauma — Adult High Risk (**9** ★) · Head Trauma — Pediatric under 2 (**8** ★) · Head Trauma — Pediatric 2 to 18 (**7** ★) · Sinusitis — Complicated Orbital (**8** ★) · Neck Mass (Adult) — Persistent Adult (**8** ★) · Neck Pain — Trauma High Risk (**9** ★) · Thoracic Back Pain — Red Flags (**7**) · Thoracic Back Pain — Post Trauma (**8** ★) · Low Back Pain — Cancer History (**7**) · Low Back Pain — Post Trauma (**8** ★) · Acute Chest Pain — Suspected ACS (**7**) · Acute Chest Pain — PE Intermediate Pretest (**8**) · Acute Chest Pain — PE High Pretest (**9**) · Chronic Cough — With Hemoptysis Weight Loss (**8** ★) · Hemoptysis — Minor (**8** ★) · Hemoptysis — Massive (**8**) · Lung Cancer Screening — Eligible 50 80 20 Pack Years (**9** ★) · RLQ Pain / Suspected Appendicitis — Adult CT (**9** ★) · Epigastric Pain / Pancreatitis — Severe Or Suspected Complication (**7**) · Suspected Small Bowel Obstruction — Standard (**9** ★) · Suspected Diverticulitis — Uncomplicated (**8** ★) · Blunt Ab-

dominal Trauma — Hemodynamically Stable (9 ★) · Renal Colic — Standard Low Dose CT (8 ★) · Hematuria — Microscopic (7 ★) · Hematuria — Gross (8 ★) · Hip Pain — Acute Trauma (7)

CTA — CT ANGIOGRAPHY

Acute Headache — Thunderclap (7) · Acute Headache — Neuro Deficit (7) · Suspected Stroke — Acute under 4 5h (9) · Suspected Stroke — Wake Up Stroke (8) · Suspected Stroke — TIA (8) · Acute Chest Pain — Suspected ACS (8 ★) · Acute Chest Pain — PE Intermediate Pretest (8 ★) · Acute Chest Pain — PE High Pretest (9 ★) · Acute Chest Pain — Aortic Dissection (9 ★) · Hemoptysis — Minor (7) · Hemoptysis — Massive (9 ★) · Stable Chest Pain — Ccta Pathway (8 ★) · Stable Chest Pain — Stress Imaging Pathway (7) · Carotid Stenosis Workup — Symptomatic (8) · AAA Screening / Surveillance — Surveillance Known AAA (7)

CT (C+) — CT WITH IV CONTRAST

Acute Headache — Fever Immunocompromised (7) · Sinusitis — Complicated Orbital (8) · Neck Mass (Adult) — Persistent Adult (8) · Neck Mass (Adult) — With Red Flags (9 ★) · Acute Chest Pain — Aortic Dissection (9) · RLQ Pain / Suspected Appendicitis — Adult CT (9) · Epigastric Pain / Pancreatitis — Severe Or Suspected Complication (8 ★) · Suspected Small Bowel Obstruction — Standard (9) · Suspected Diverticulitis — Uncomplicated (8) · Suspected Diverticulitis — Complicated (9 ★) · Blunt Abdominal Trauma — Hemodynamically Stable (9) · Hematuria — Microscopic (7) · Hematuria — Gross (8)

MRI — MRI, NON-CONTRAST

Acute Headache — Neuro Deficit (9 ★) · Acute Headache — New over 50 (8 ★) · Suspected Stroke — Acute under 4 5h (8) · Suspected Stroke — Wake Up Stroke (9 ★) · Suspected Stroke — TIA (9 ★) · Seizure — First Seizure (9 ★) · Seizure — Breakthrough (8 ★) · Sinusitis — Complicated Orbital (7) · Neck Mass (Adult) — Persistent Adult (7) · Neck Pain — Myelopathy Signs (9 ★) · Thoracic Back Pain — Red Flags (9 ★) · Low Back Pain — Radiculopathy over 6wk (8 ★) · Low Back Pain — Cauda Equina (9 ★) · Low Back Pain — Cancer History (9 ★) · Low Back Pain — Osteoporosis Fragility (9 ★) · Low Back Pain — Post Trauma (8) · New Heart Failure — Initial Evaluation (7) · RLQ Pain / Suspected Appendicitis — Pregnant US MRI (8) · RLQ Pain / Suspected Appendicitis — Pediatric US First (8) · Pelvic Pain (Female) — Post Menopausal Bleeding (7) · Renal Colic — Pregnant US First (7) · Shoulder Pain — Chronic Rotator Cuff MRI (8 ★) · Shoulder Pain — Chronic Rotator Cuff US (8) · Knee Pain — Meniscal Ligament Post Xr (8 ★) · Hip Pain — Occult Fracture Negative Xr (9 ★) · Breast Cancer Screening — High Risk MRI Supplement (8 ★)

MRI (C+) — MRI WITH CONTRAST

Acute Headache — Fever Immunocompromised (9 ★) · Low Back Pain — Infection IVDU (9 ★)

MRA — MR ANGIOGRAPHY

Acute Headache — Neuro Deficit (7) · Suspected Stroke — Acute under 4 5h (8) · Suspected Stroke — TIA (8) · Acute Chest Pain — Aortic Dissection (8) · Carotid Stenosis Workup — Symptomatic (8)

US — ULTRASOUND

Thyroid Nodule — Initial Characterization (9 ★) · New Heart Failure — Initial Evaluation (9 ★) · Preoperative Cardiac Clearance — High Risk Surgery Symptoms (8 ★) · RLQ Pain / Suspected Appendicitis — Pregnant US MRI (9 ★) · RLQ Pain / Suspected Appendicitis — Pediatric US First (9 ★) · RUQ Pain / Suspected Cholecystitis — US First (9 ★) · Blunt Abdominal Trauma — Hemodynamically Stable (7) · Blunt Abdominal Trauma — Hemodynamically Unstable (9 ★) · Pelvic Pain (Female) — US First (9 ★) · Pelvic Pain (Female) — Ectopic Workup (9 ★) · Pelvic Pain (Female) — Post Menopausal Bleeding (8 ★) · Renal Colic — Pregnant US First (9 ★) · Renal Colic — Young Recurrent US (8 ★) · Scrotal Pain — Standard Doppler (7) · Shoulder Pain — Chronic Rotator Cuff MRI (7) · Shoulder Pain — Chronic Rotator Cuff US (8 ★) · AAA Screening / Surveillance — Screening Eligible (9 ★) ·

AAA Screening / Surveillance — Surveillance Known AAA (9 ★) · Palpable Breast Lump — under 30 US First (9 ★) · Palpable Breast Lump — 30 + Mammo US (9)

US DOPPLER — ULTRASOUND, DOPPLER/DUPLEX

Scrotal Pain — Standard Doppler (9 ★) · Scrotal Pain — Suspected torsion (9 ★) · Suspected DVT — Standard Doppler (9 ★) · Carotid Stenosis Workup — Symptomatic (9 ★) · Carotid Stenosis Workup — Asymptomatic Screening (7 ★)

NM — NUCLEAR MEDICINE

Acute Chest Pain — Suspected ACS (8) · Stable Chest Pain — Ccta Pathway (7) · Stable Chest Pain — Stress Imaging Pathway (8 ★) · Preoperative Cardiac Clearance — High Risk Surgery Symptoms (7)

PET-CT — PET-CT

Neck Mass (Adult) — With Red Flags (7)

FLUORO — FLUOROSCOPY

No usually-appropriate (7-9) variant in the current knowledge base.

MAMMO — MAMMOGRAPHY

Breast Cancer Screening — Average Risk 40 + (9 ★) · Breast Cancer Screening — High Risk MRI Supplement (9) · Palpable Breast Lump — 30 + Mammo US (9 ★)

DEXA — DEXA DENSITOMETRY

Low Back Pain — Osteoporosis Fragility (7)

Appendix I — Methodology & change control

- **Version.** Knowledge Matrix v1.0.0, stamped into every score for audit.
- **Determinism.** Identical input -> identical score, factors, and matrix stamp; rules path runs without ML.
- **Integrity checks.** Unique scenario IDs and unique, internally consistent evidence slugs enforced at load.
- **Source of truth.** `lib/aiie/scoring-engine.ts` and `lib/aiie/knowledge-matrix/` ; this document is generated from that data and is subordinate to it.
- **Change process.** Rating changes are matrix-version-bumped and regression-tested before release.

Appendix J — Coverage statistics

36 SCENARIOS	93 VARIANTS	1,302 MODALITY RATINGS	16 NO-IMAGING VARIANTS	93 PREFERRED-STUDY TAGS
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Region	Scenarios	Variants
Head / Brain	4	17
Head, Face & Neck	3	5
Cervical Spine	1	4
Thoracic Spine	1	3
Lumbar Spine	1	7
Chest	4	11
Cardiac	3	5
Abdomen	6	11
Pelvis	1	3
GU / Renal	3	7
Upper Extremity MSK	1	3
Lower Extremity MSK	3	8
Vascular	3	5
Breast	2	4

Appendix K — Glossary

AIIE	ARKA Imaging Intelligence Engine - the appropriateness scoring engine documented here .
AIIE score	ARKA's 1-9 appropriateness output (1-3 red, 4-6 amber, 7-9 green).
Non-Device CDS	Clinical decision support meeting FD&C §520(o)(1)(E): no image analysis, transparent basis, clinician retains the decision.
SHAP-style factor	A signed, bounded per-factor contribution to the score, shown for transparency.
RAND/UCLA method	Consensus appropriateness methodology underpinning the 1-9 construct.
GRADE	Evidence-to-decision grading framework informing rationale transparency.
Clinical Knowledge Matrix	ARKA's deterministic scenario->variant->modality rating knowledge base.
Match tier	How specifically the matrix matched an order (exact / scenario / region / conservative).
Evidence slug	Stable identifier linking a variant's rating set to its evidence metadata.
MNAI	Medical Necessity Alignment Index - ICD-10/CPT alignment to payer policy (0-100, green/amber/red).
Denial-risk proxy	Payer-facing mirror of the clinical score: 10 - AIIE score.
Gold card	Provider status that auto-approves prior authorization in ARKA-INS.
Red flag	A high-acuity clinical discriminator that selects urgent variants and escalates the score.
Radiation tier	Relative ionising-radiation burden (0 none -> 4 high).
VFSS / FEES	Video-fluoroscopic swallow study / fiberoptic endoscopic evaluation of swallowing.

End of document. Generated from ARKA source at Knowledge Matrix v1.0.0 on June 10, 2026. Where this document and the code disagree, the code governs.